INCH POUND

MIL-M-38510/15B
14 April 2005
SUPERSEDING
MIL-M-38510/15A
27 July 1979

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, BISTABLE LATCHES, MONOLITHIC SILICON

Inactive for new design after 6 September 1996.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF 38535

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic, silicon, TTL, 4-bit bistable latch microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.4).

1.2 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-38535, and as specified herein.

1.2.1 Device types. The device types are as follows:

Device type	Circuit
01	4-bit latch, complementary outputs
02	4-bit latch
03	Dual 4-bit latch
04	4-bit latch, master reset

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines are as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
A	GDFP5-F14 or CDFP6-F14	14	Flat pack
B	GDFP4-F14	14	Flat pack
C	GDIP1-T14 or CDIP2-T14	14	Dual-in-line
D	GDFP1-F14 or CDFP2-F14	14	Flat pack
E	GDIP1-T16 or CDIP2-T16	16	Dual-in-line
F	GDFP2-F16 or CDFP3-F16	16	Flat-pack
J	GDIP1-T24 or CDIP2-T24	24	Dual-in-line
K	GDFP2-F24 or CDFP3-F24	24	Flat-pack
L	GDIP3-T24 or CDIP4-T24	24	Dual-in-line
Z	GDFP7-F24 or CDFP8-F24	24	Flat-pack

Comments, suggestions, or questions on this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAS, P. O. Box 3990, Columbus, OH 43218-3990, or emailed to bipolar@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 5962

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -12 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation per gate, (P_D) 1/	
Device types 01 and 02	280 mW dc
Device type 03	630 mW dc
Device type 04	325 mW dc
Lead temperature (soldering 10 seconds)	300°C
Thermal resistance, junction-to-case (θ_{JC})	(See MIL-STD-1835)
Junction temperature (T_J) 2/	175°C

1.4 Recommended operating conditions.

Supply voltage	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage	2.0 V dc
Maximum low level input voltage	0.8 V dc
Normalized fanout (each output) 3/	5 maximum
Case operating temperature range (T_C)	-55°C to 125°C
Setup time, t($_{SETUP}$) data to output	
Types 01 and 02	25 ns minimum
Setup time t($_{SETUP}$), type 03	
Data to enable (high)	10 ns minimum
Data to enable (low)	16 ns minimum
Setup time t($_{SETUP}$), type 04	
Data to enable (high)	5 ns minimum
Data to enable (low)	25 ns minimum
Setup time, t($_{SETUP}$) data to set (high)	
Type 04	8 ns minimum
Input hold time t($_{HOLD}$), types 01 and 02	
Clock to data	5 ns minimum
Data to clock	30 ns minimum
Input hold time t($_{HOLD}$), type 03	
Data to enable (high)	0 ns maximum
Data to enable (low)	4 ns maximum
Input hold time t($_{HOLD}$), type 04	
Data to enable (high)	0 ns maximum
Data to enable (low)	7 ns maximum
Input hold time t($_{HOLD}$), type 04	
Data to set (low)	8 ns maximum

2.0 APPLICABLE DOCUMENT

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

1/ Must withstand the added P_D due to short circuit condition (e.g. I_{OS}) at one output for 5 seconds duration.
2/ Maximum junction temperature should not be exceeded except in accordance with allowable short duration burn-in screening condition in accordance with MIL-PRF-38535.
3/ Device will fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.3).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.3.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.3.3 Logic diagrams. The logic diagrams shall be as specified on figure 3.

3.3.4 Schematic circuit. The schematic circuit shall be maintained by the manufacturer and made available to the qualifying activity and the preparing activity upon request.

3.3.5 Case outlines. Case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. Lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. The electrical performance characteristics are as specified in table 1 and apply over the full recommended case operating temperature range, unless otherwise specified.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 5 (see MIL-PRF-38535, appendix A).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5V; I_{OH} = -400µA V_{IN} = 0.8V	01, 02	2.4		V
		I_{OH} = -800µA	03, 04	2.4		V
Low level output voltage	V_{OL}	V_{CC} = 4.5V; I_{OL} = 16 mA	01,02,04		0.4	V
		V_{IN} = 2.0V I_{OL} = 14.4 mA	03		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; V_{IN} = -12 mA	All		-1.5	V
Low level input current, data	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = 0.4 V	01, 02	-1.4	-3.2	mA
			03 4/	-0.5	-2.4	mA
			04	-0.7	-2.7	mA
Low level input current	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = 0.4 V 1/				
Clock			01, 02	-2.8	-6.4	mA
Master reset and enable			03, 04	-0.7	-1.6	mA
Set			04	-0.7	-1.6	mA
High level input current, data	I_{IH1}	V_{CC} = 5.5 V; V_{IN} = 2.4 V 1/	01, 02		80	µA
			03		40	µA
			04		60	µA
High level input current, data	I_{IH2}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 1/	01, 02		200	µA
			03, 04		100	µA
High level input current	I_{IH3}	V_{CC} = 5.5 V; V_{IN} = 2.4 V 1/				
Clock			01, 02		160	µA
Master reset and enable			03, 04		40	µA
Set			04		40	µA
High level input current	I_{IH4}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 1/				
Clock			01, 02		400	µA
Master reset and enable			03, 04		100	µA
Set			04		100	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V 2/ 3/	01, 02	-20	-57	mA
			03 5/	-10	-70	mA
			04	-20	-100	mA
Supply current	I_{CC}	V_{CC} = 5.5 V 2/; V_{IN} = 4.5 V	01, 02		46	mA
			03		106	mA
			04		55	mA
Propagation delay time for types 01 and 02						
To high level, from D input to Q output	t_{PLH1}	C_L = 50 pF ± 10%; R_L = 390 ohms ± 5%	01, 02	2	44	ns
To high level, from D input to $\overline{Q}$ output	t_{PLH2}		01	2	55	ns
To low level, from D input to Q output	t_{PHL1}		01, 02	2	38	ns
To low level, from D input to $\overline{Q}$ output	t_{PHL2}		01	2	25	ns
To low level, from clock input to Q output	t_{PHL3}		01	2	25	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
To high level, from clock input to Q output	t_{PLH3}	C_L = 50 pF ± 10%; R_L = 390 ohms ± 5%	01, 02	2	44	ns
To low level, from clock input to $\overline{Q}$ output	t_{PHL4}		01	2	25	ns
To high level from clock input to $\overline{Q}$ output	t_{PLH4}		01	2	44	ns
Propagation delay time for types 03 and 04						
To high level, from enable to output	t_{PLH1}	C_L = 50 pF ± 10%; R_L = 390 ohms ± 5%	03, 04	3	60	ns
To low level, from enable to output	t_{PHL1}		03, 04	3	40	ns
To high level, from data to output	t_{PLH2}		03, 04	3	49	ns
To low level, from data to output	t_{PHL2}		03, 04	3	37	ns
To low level, from master reset to output	t_{PHL3}		03, 04	3	37	ns
To high level, from set to output	t_{PLH4}		04	9	47	ns

1/ All unspecified inputs at 5.5 V.
2/ All unspecified inputs grounded.
3/ Not more than one output should be shorted at a time.
4/ Circuit B limits for I_{IL1} shall be -1.6 mA, maximum.
5/ Circuit B and C limits for I_{OS} shall be -20 mA minimum to -57 mA maximum.

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TABLE II. Electrical test requirements.

MIL-PRF-38535 Test requirement	Subgroups (see table III)	
	Class S Devices	Class B Devices
Interim electrical parameters	1	1
Final electrical test parameters	1*, 2, 3, 9, 10, 11	1*, 2, 3, 9
Group A test requirements	1, 2, 3, 9, 10, 11	1, 2, 3, 9,
Group B electrical test parameters when using the method 5005 QCI option	1, 2, 3	N/A
Groups C end point electrical parameters	1, 2, 3	1, 2, 3
Additional electrical subgroups for Group C periodic inspections	N/A	10, 11
Group D end point electrical parameters	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1.

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.3 Screening. Screening shall be in accordance with MIL-PRF-38535 and shall be conducted on all devices prior to qualification and conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535.

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4.4 Technology Conformance Inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End point electrical parameters shall be as specified in table II *herein*.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

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Terminal number	Device type 01	Device type 02	Device type 03	Device type 04
	Cases E and F	Cases A, B, C and D	Cases J, K, L and Z	Cases E and F
1	1 $\overline{Q}$	1D	$\overline{MR}$	$\overline{E}$
2	1D	2D	$\overline{E}$ 0	$\overline{S}$ 0
3	2D	CLOCK 3-4	$\overline{E}$ 1	D0
4	CLOCK 3-4	V_{CC}	D0	D1
5	V_{CC}	3D	Q0	$\overline{S}$ 2
6	3D	4D	D1	D2
7	4D	NC	Q1	D3
8	4 $\overline{Q}$	4Q	D2	GND
9	4Q	3Q	Q2	$\overline{MR}$
10	3Q	NC	D3	Q3
11	3 $\overline{Q}$	GND	Q3	$\overline{S}$ 3
12	GND	CLOCK 1-2	GND	Q2
13	CLOCK 1-2	2Q	$\overline{MR}$	Q1
14	2 $\overline{Q}$	1Q	$\overline{E}$ 0	$\overline{S}$ 1
15	2Q		$\overline{E}$ 1	Q0
16	1Q		D0	V_{CC}
17			Q0	
18			D1	
19			Q1	
20			D2	
21			Q2	
22			D3	
23			Q3	
24			V_{CC}	

Figure 1. Terminal connections.

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Device types 01 and 02

Truth table	
tn	tn+1
D	Q
H	H
L	L

NOTES:
1. tn = bit time before clock negative-going transition.
2. tn+1 = bit time after clock negative-going transition.

Device type 03

Truth table					
$\overline{MR}$	$\overline{E}$ 0	$\overline{E}$ 1	D	Qn	Operation
H	L	L	L	L	Data entry
H	L	L	H	H	Data entry
H	L	H	X	Qn-1	Hold
H	H	L	X	Qn-1	Hold
H	H	H	X	Qn-1	Hold
L	X	X	X	L	reset

X = Don't care
L = Low voltage level
H = High voltage level
Qn-1 = Previous output state
Qn = Present output state

Device type 04

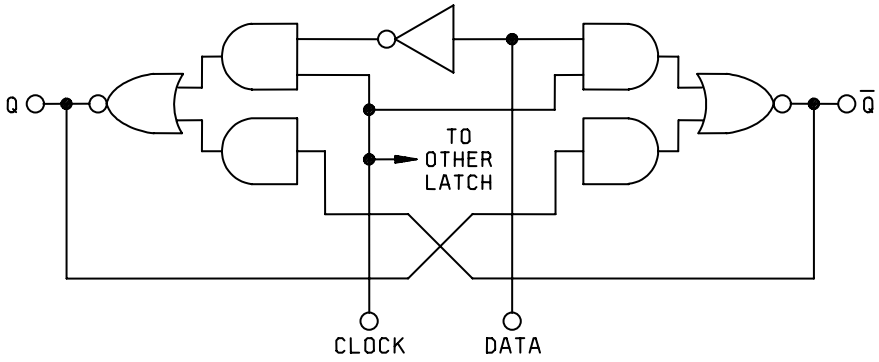
Truth table					
$\overline{MR}$	$\overline{E}$	D	$\overline{S}$	Qn	Operation
H	L	L	L	L	D mode
H	L	H	L	H	
H	H	X	X	Qn-1	
H	L	L	L	L	R/S mode
H	L	H	L	H	
H	L	L	H	L	
H	L	H	H	Qn-1	
H	H	X	X	Qn-1	
L	X	X	X	L	Reset

X = Don't care
L = Low voltage level
H = High voltage level
Qn-1 = Previous output state
Qn = Present output state

Figure 2. Truth tables.

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Device type 01



Device type 02

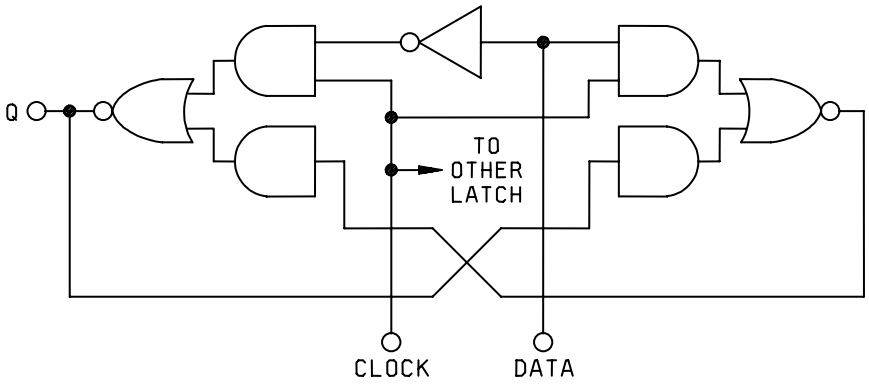


Figure 3. Logic diagrams.

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Device type 03

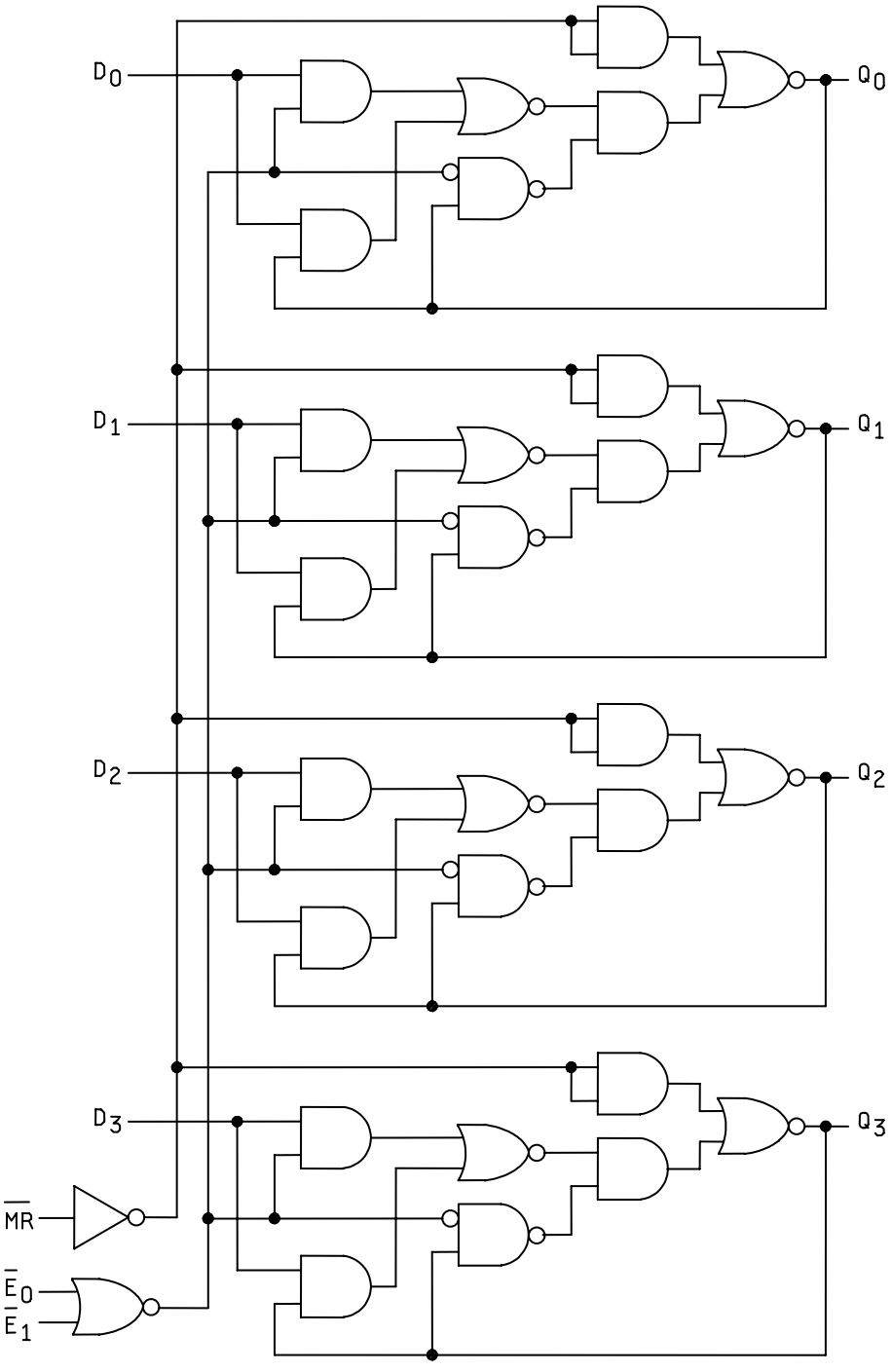


Figure 3. Logic diagrams – Continued.

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Device type 04

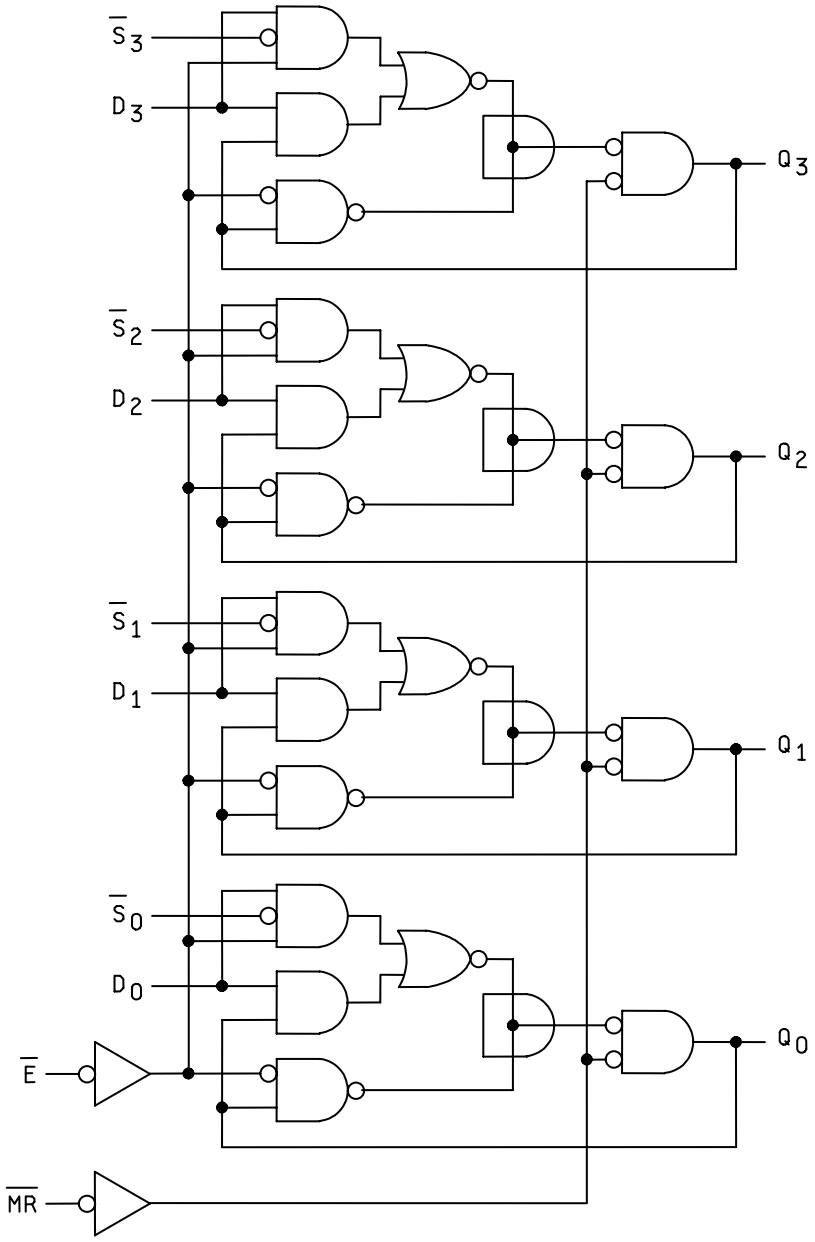
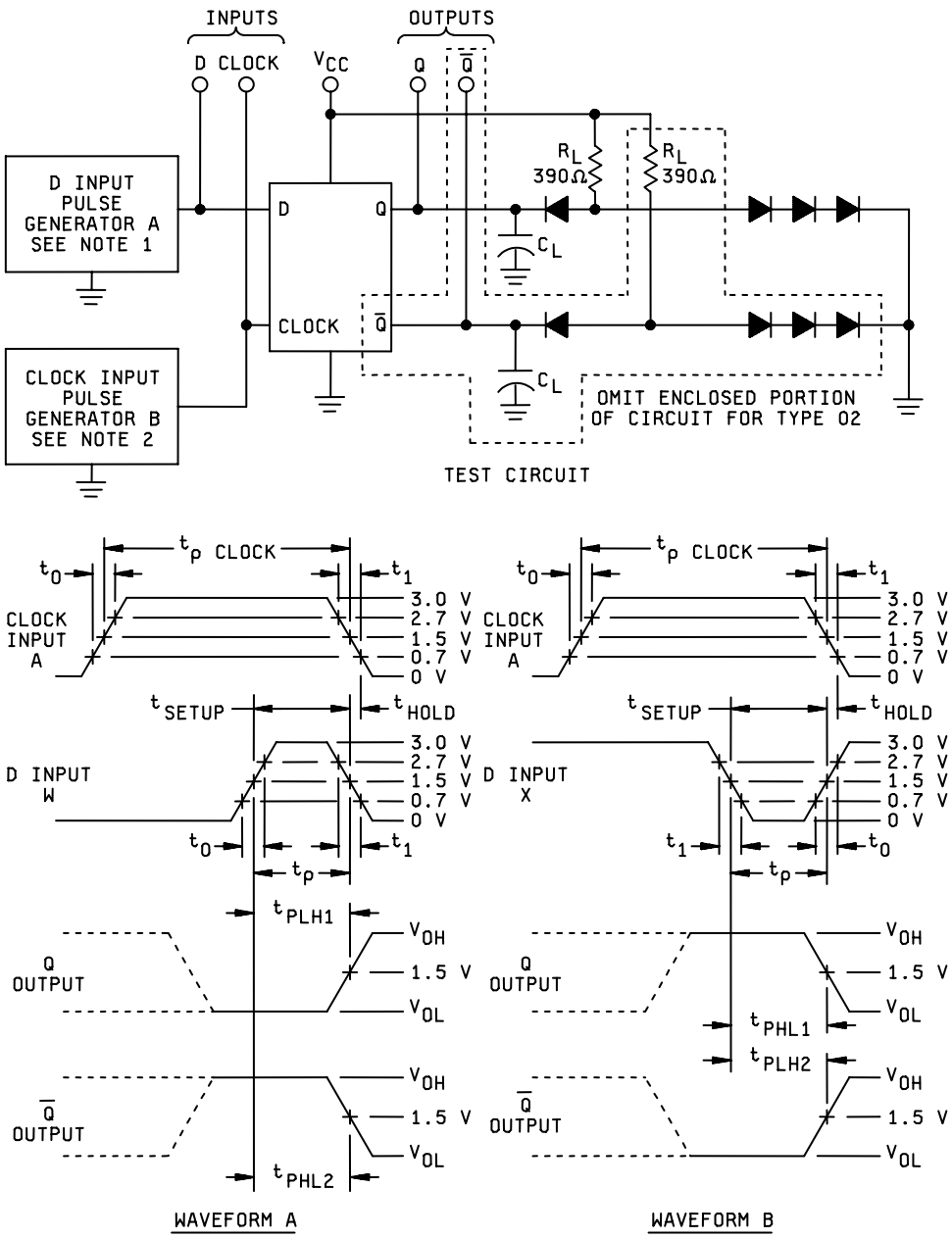


Figure 3. Logic diagrams – Continued.

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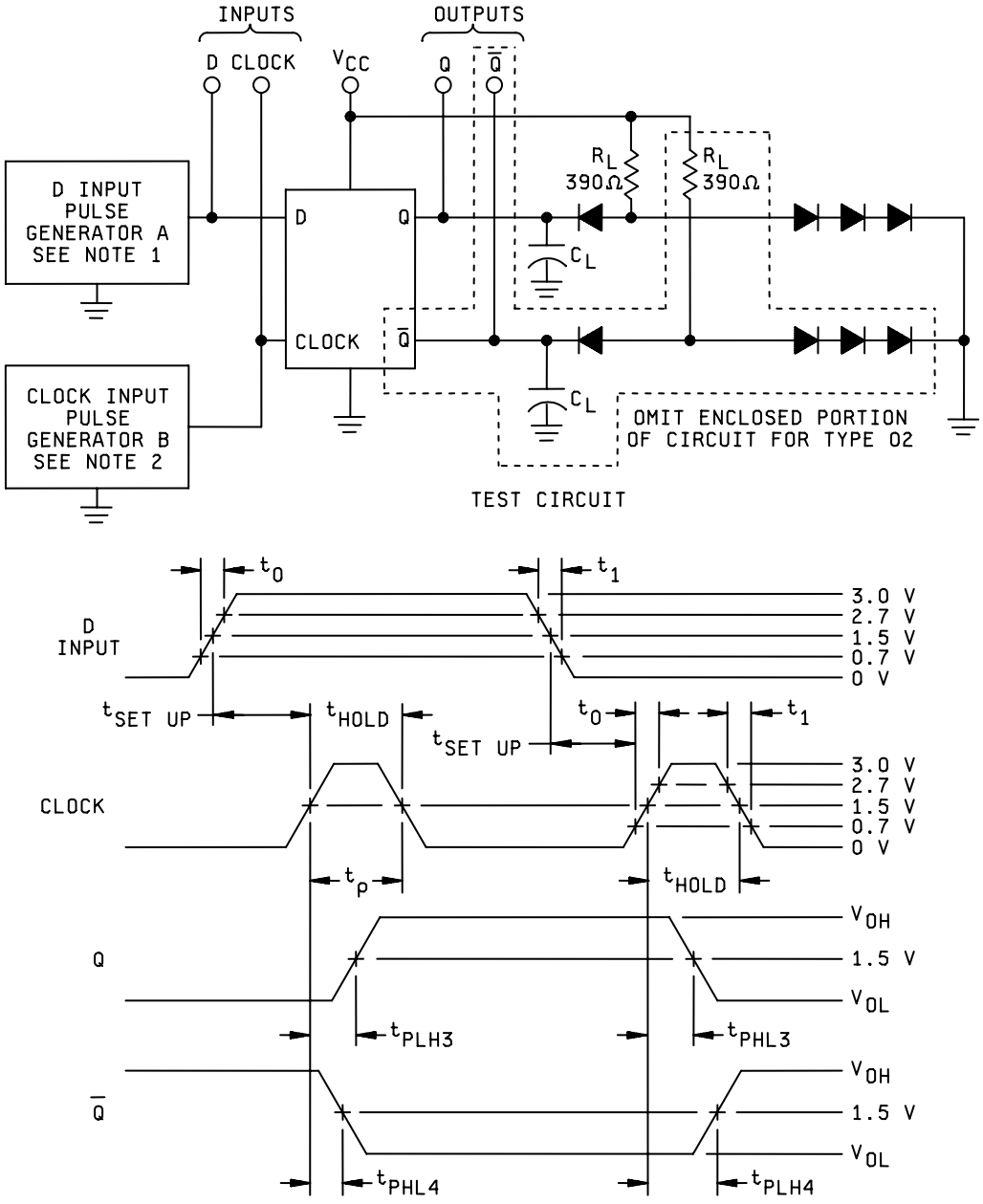


NOTES:

1. The D input pulse generator has the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, t_P = 30 ns, $t_{(SETUP)}$ = 25 ns, $t_{(HOLD)}$ = 5 ns, PRR = 1 MHz and $Z_{OUT} \approx$ 50 Ω.
2. The clock pulse generator has the following characteristics: V_{GEN} = 3 V minimum, $t_1 = t_0 \leq 10$ ns, $t_{P(CLOCK)}$ = 500 ns, and PRR = 1 MHz.
3. Prior to testing the device shall be precondition to a high logic level for waveform A and to a low logic level for waveform B.
4. Each latch is tested separately.
5. C_L = 50 pF, which includes probe and jig capacitance.
6. R_L = 390 Ω ±5%.
7. The $\overline{Q}$ waveforms are not applicable to type 02.
8. All diodes are 1N3064 or equivalent.

Figure 4. Switching test circuit and data to output waveforms for device types 01 and 02.

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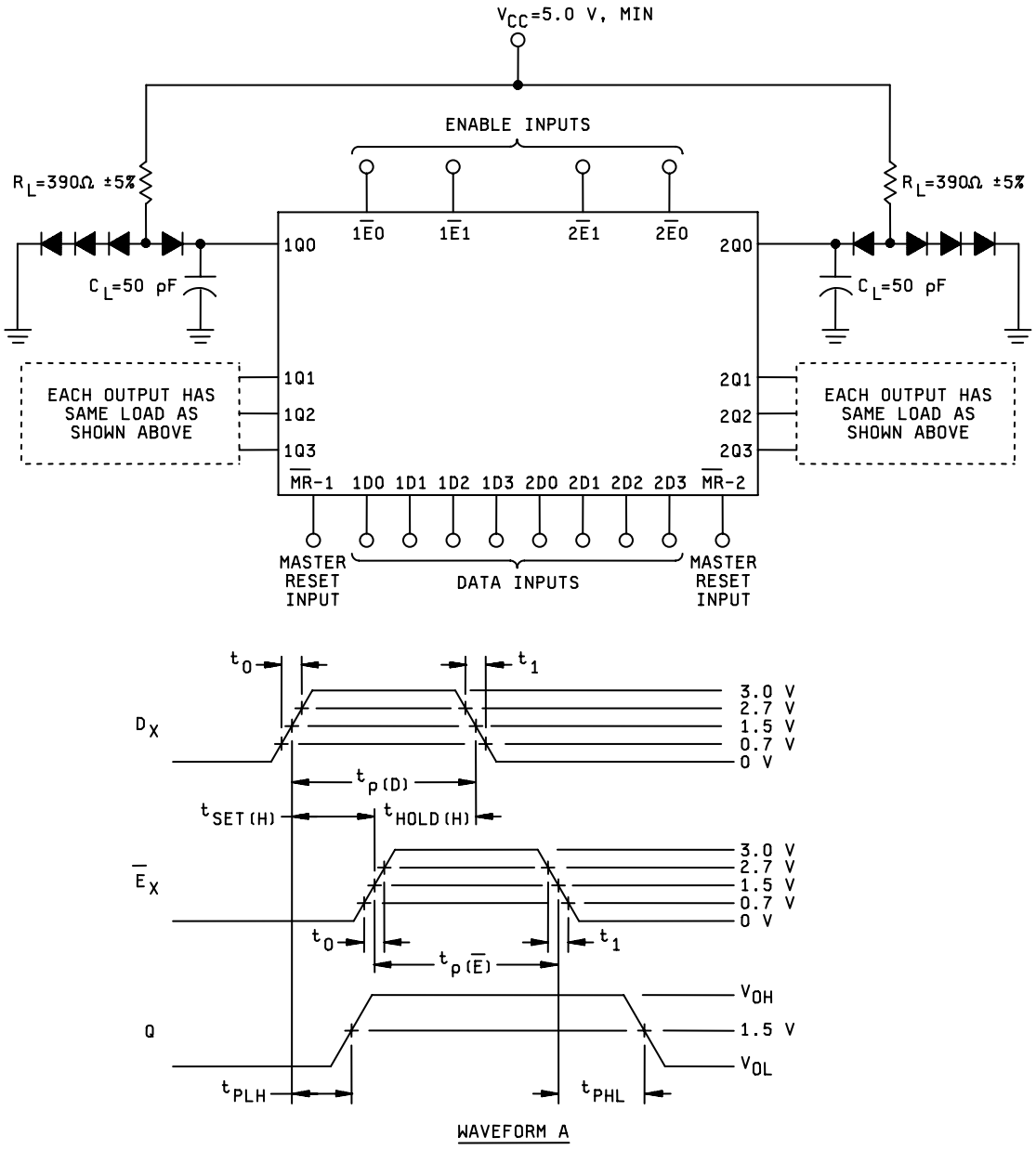


NOTES:

1. The D input pulse generator has the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, PRR = 500 kHz at 50 % duty cycle. For subgroups 7 and 8, PRR ≤ 25 kHz at 50 % cuty cycle and PRR (D input) = 1/2 PRR (clock).
2. The clock pulse generator has the following characteristics: V_{GEN} = 3 V minimum, $t_1 = t_0 \leq 10$ ns, $t_{(HOLD)} = t_{P(CLOCK)}$ = 30 ns, $t_{(SETUP)}$ = 25 ns and PRR = 1 MHz. For subgroups 7 and 8, PRR ≤ 50 kHz.
3. Each latch is tested separately.
4. C_L = 50 pF, which includes probe and jig capacitance.
5. R_L = 390 Ω ±5%.
6. The $\overline{Q}$ waveforms are not applicable to device type 02.
7. All diodes are 1N3064 or equivalent.

Figure 5. Switching test circuit and clock to output waveforms for device types 01 and 02.

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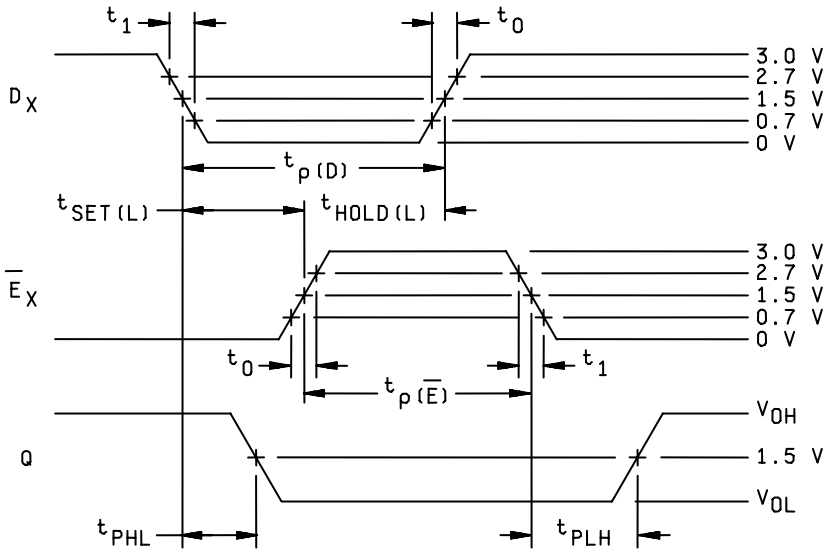


NOTES:

1. The data inputs have the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, $t_{P(D)}$ = 10 ns, PRR = 1 MHz.
2. The enable inputs have the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, $t_{SET(H)}$ = 10 ns, $t_{HOLD(H)}$ = 0 ns, $t_{P(E)}$ = 20 ns, PRR = 1 MHz and $Z_{OUT} \approx 50\Omega$.
3. Each latch is tested separately.
4. C_L = 50 pF, which includes probe and jig capacitance.
5. R_L = 390 Ω ±5%.
6. All diodes are 1N3064 or equivalent.

Figure 6. Switching time test circuit and waveforms for device type 03.

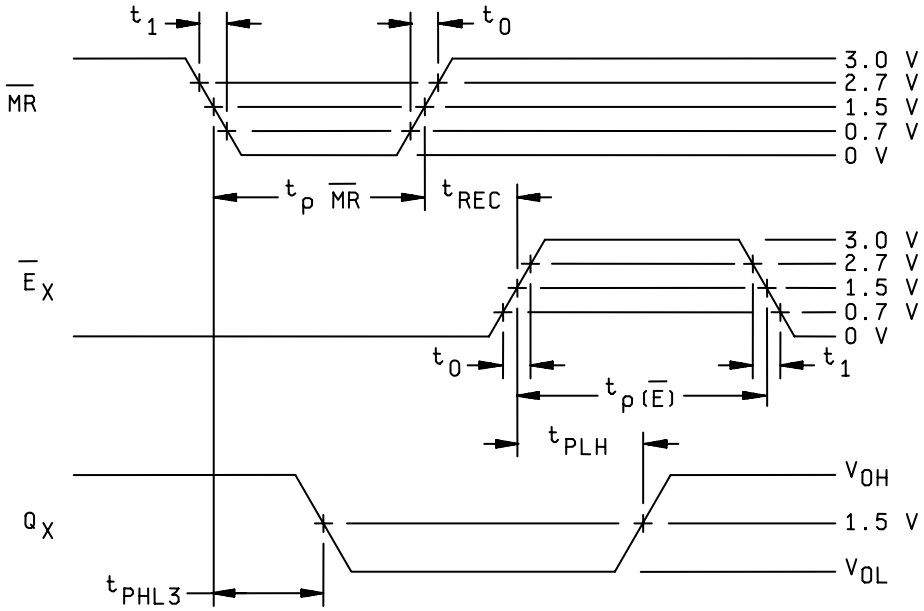
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WAVEFORM B

NOTES:

1. The master reset inputs have the following characteristics: V_{GEN} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{P(D)}$ = 26 ns and PRR = 1 MHz.
2. The enable inputs have the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, $t_{SET(L)}$ = 16 ns, $t_{HOLD(L)}$ = 10 ns, $t_{P(\overline{E})}$ = 20 ns, PRR = 1 MHz and $Z_{OUT} \approx 50\Omega$.



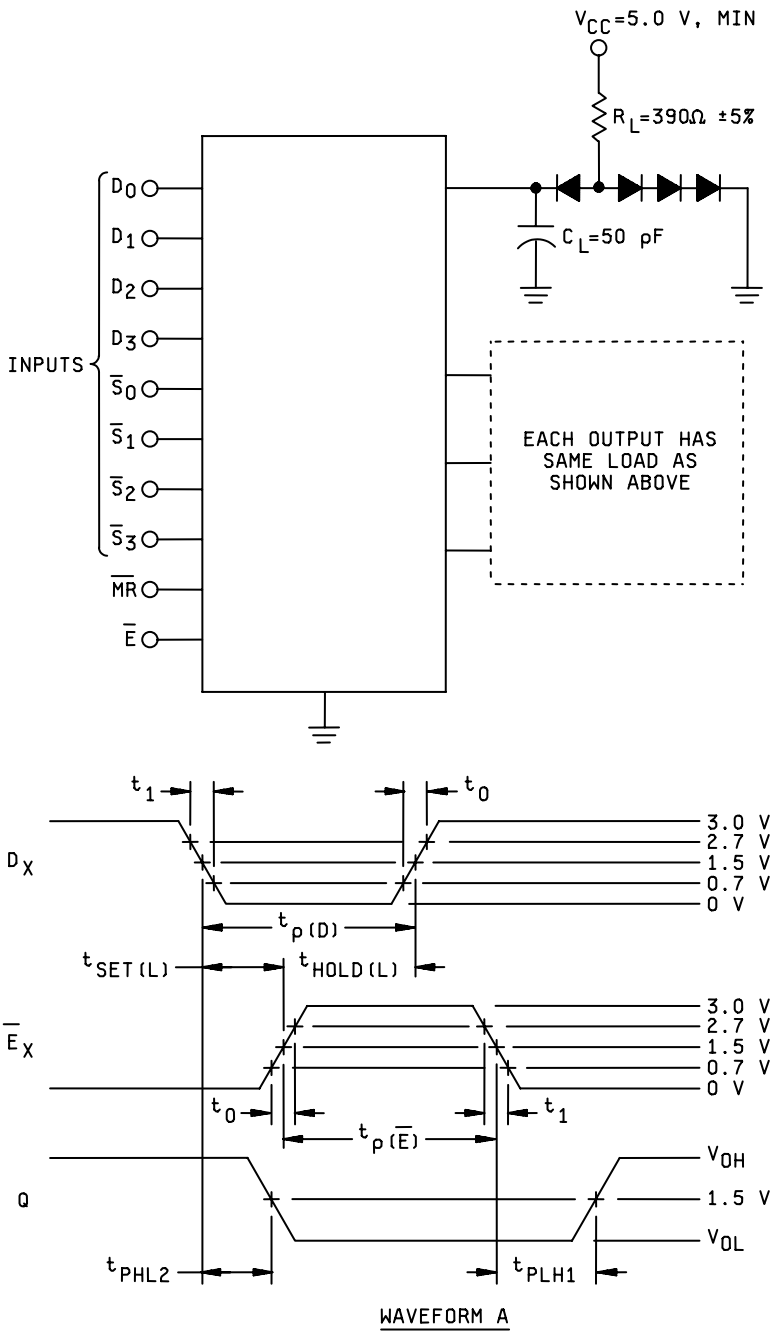
WAVEFORM C

NOTES:

1. The master reset inputs have the following characteristics: V_{GEN} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{P(\overline{MR})}$ = 20 ns and PRR = 1 MHz.
2. The enable inputs have the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, t_{REC} = 10 ns, $t_{P(\overline{E})}$ = 20 ns, PRR = 1 MHz and $Z_{OUT} \approx 50\Omega$.

Figure 6. Switching time test circuit and waveforms for device type 03.

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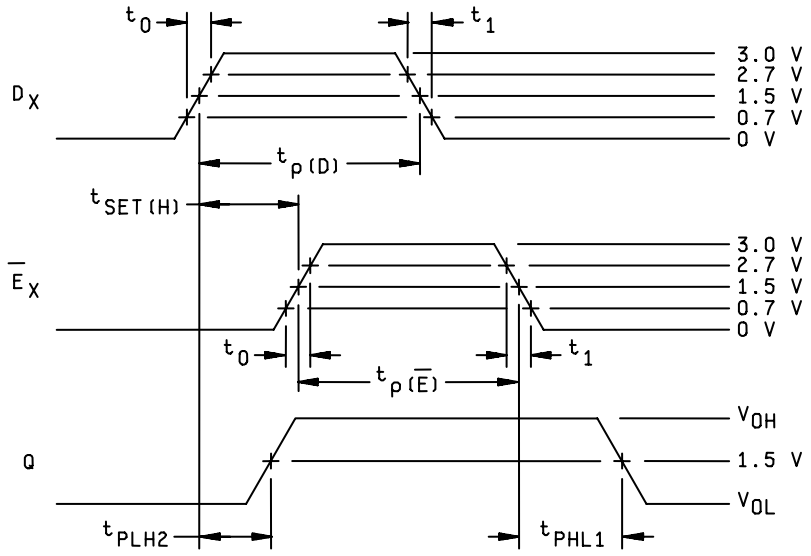
WAVEFORM A

NOTES:

1. The data inputs have the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, $t_{P(D)}$ = 32 ns, PRR = 1 MHz.
2. The enable input has the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, $t_{P(E)}$ = 20 ns, $t_{SET(L)}$ = 25 ns, $t_{HOLD(L)}$ = 7 ns, PRR = 1 MHz and $Z_{OUT} \approx 50\Omega$.
3. Each latch is tested separately.
4. C_L = 50 pF, which includes probe and jig capacitance.
5. R_L = 390 Ω ±5%.
6. All diodes are 1N3064 or equivalent.

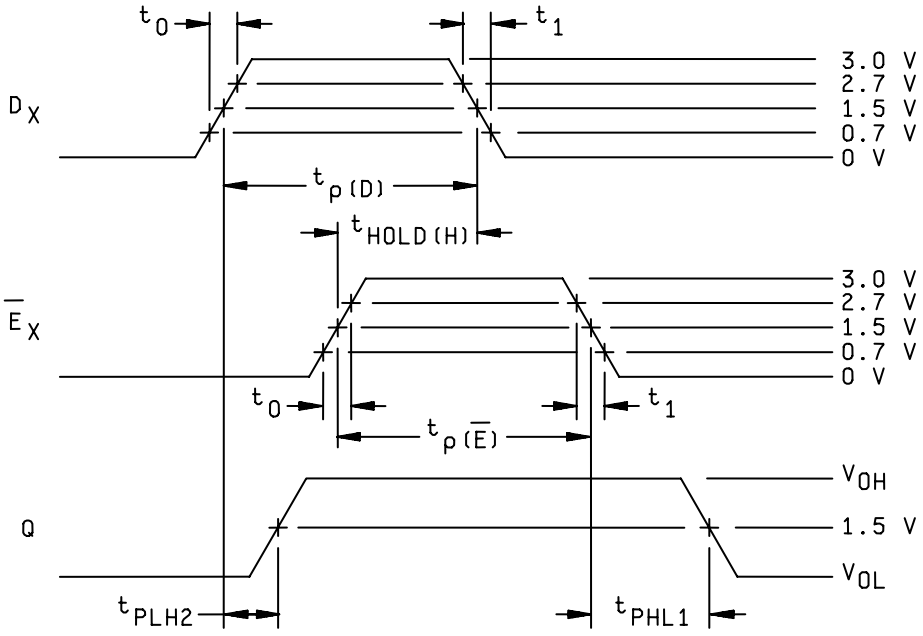
Figure 7. Switching time test circuit and waveforms for device type 04.

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WAVEFORM B

NOTES:
1. The data inputs have the following characteristics: V_{GEN} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{P(D)}$ = 20 ns and PRR = 1 MHz.
2. The enable input has the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, $t_{P(\overline{E})}$ = 20 ns, $t_{SET(H)}$ = 5 ns, PRR = 1 MHz and $Z_{OUT} \approx 50\Omega$.

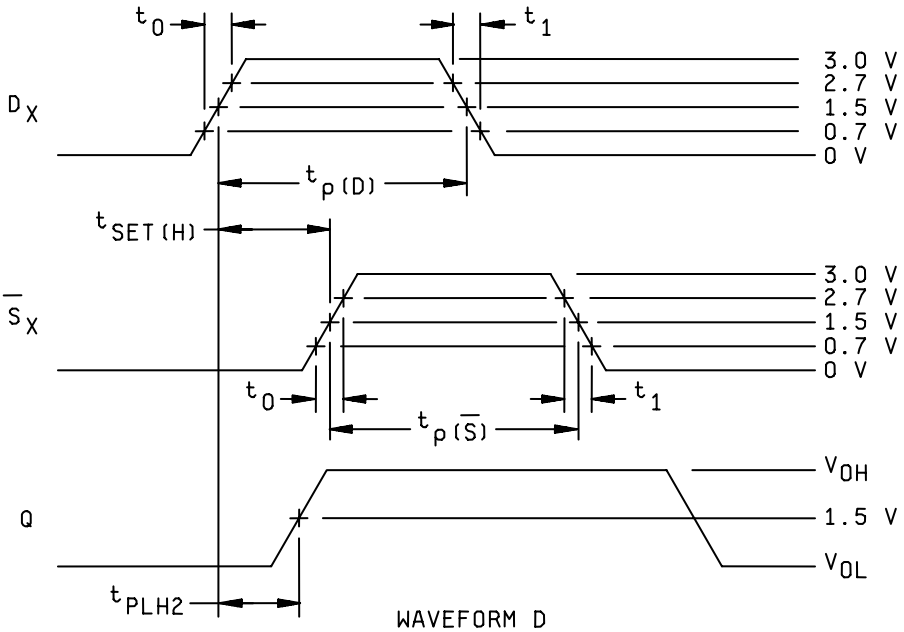


WAVEFORM C

NOTES:
1. The data inputs have the following characteristics: V_{GEN} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{P(D)}$ = 20 ns and PRR = 1 MHz.
2. The enable inputs have the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, $t_{P(\overline{E})}$ = 20 ns, $t_{HOLD(H)}$ = 0 ns, PRR = 1 MHz and $Z_{OUT} \approx 50\Omega$.

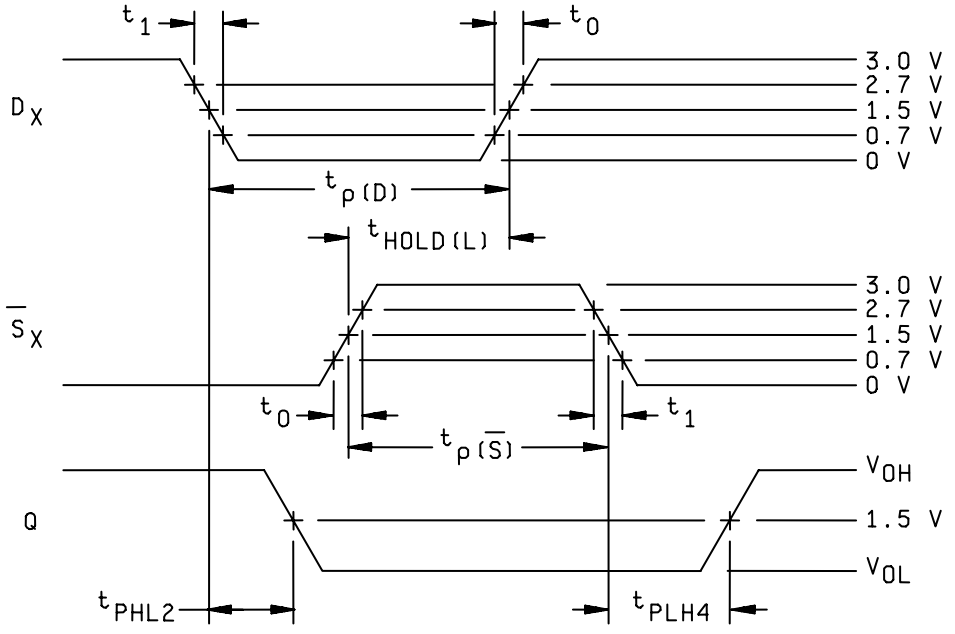
Figure 7. Switching time test circuit and waveforms for device type 04.

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WAVEFORM D

NOTES:
1. The data inputs have the following characteristics: V_{GEN} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{P(D)}$ = 20 ns and PRR = 1 MHz.
2. The set inputs have the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, $t_{P(\overline{S})}$ = 20 ns, $t_{SET(H)}$ = 8 ns, PRR = 1 MHz and $Z_{OUT} \approx 50\Omega$.

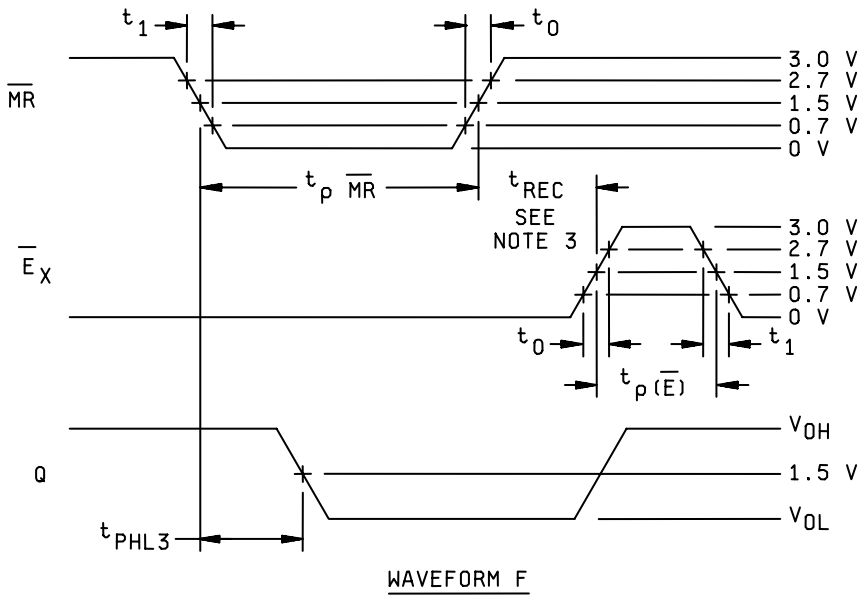


WAVEFORM E

NOTES:
1. The data inputs have the following characteristics: V_{GEN} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{P(D)}$ = 20 ns and PRR = 1 MHz.
2. The set inputs have the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, $t_{P(\overline{S})}$ = 20 ns, $t_{HOLD(L)}$ = 8 ns, PRR = 1 MHz and $Z_{OUT} \approx 50\Omega$.

Figure 7. Switching time test circuit and waveforms for device type 04.

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WAVEFORM F

NOTES:

1. The master reset input has the following characteristics: V_{GEN} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{P(\overline{MR})}$ = 18 ns and PRR = 1 MHz.
2. The enable inputs have the following characteristics: V_{GEN} = 3 V, $t_1 = t_0 \leq 10$ ns, $t_{P(\overline{E})}$ = 20 ns, t_{REC} = 0 ns, PRR = 1 MHz and $Z_{OUT} \approx 50\Omega$.
3. Recovery time is the minimum time that enable must remain low after the master reset transition from low to high in order for the latch to recognize and store high data.

Figure 7. Switching time test circuit and waveforms for device type 04.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	1 $\overline{Q}$	1D	2D	CLOCK 3-4	V_{CC}	3D	4D	4 $\overline{Q}$	4Q	3Q	3 $\overline{Q}$	GND	CLOCK 1-2	2 $\overline{Q}$	2Q	1Q		Min	Max	Unit
1	V_{OH}	3006	1	-0.4 mA	0.8 V			4.5 V							GND	2.0 V				1 $\overline{Q}$	2.4		V
T_C = 25°C	"	"	2		2.0 V			"							"	"			-0.4 mA	1Q	"		"
	"	"	3			0.8 V		"							"	"	-0.4 mA			2 $\overline{Q}$	"		"
	"	"	4			2.0 V		"							"	"		-0.4 mA		2Q	"		"
	"	"	5				2.0 V	"	0.8 V					-0.4 mA	"					3 $\overline{Q}$	"		"
	"	"	6				"	"	2.0 V				-0.4 mA		"					3Q	"		"
	"	"	7				"	"		0.8 V	-0.4 mA				"					4 $\overline{Q}$	"		"
	"	"	8				"	"		2.0 V		-0.4 mA			"					4Q	"		"
	V_{OL}	3007	9	16 mA	2.0 V			"							"	2.0 V				1 $\overline{Q}$		0.4	"
	"	"	10		0.8 V			"							"	"			16 mA	1Q		"	"
	"	"	11			2.0 V		"							"	"	16 mA			2 $\overline{Q}$		"	"
	"	"	12			0.8 V		"							"	"		16 mA		2Q		"	"
	"	"	13				2.0 V	"	2.0 V					16 mA	"					3 $\overline{Q}$		"	"
	"	"	14				"	"	0.8 V				16 mA		"					3Q		"	"
	"	"	15				"	"		2.0 V	16 mA				"					4 $\overline{Q}$		"	"
	"	"	16				"	"		0.8 V		16 mA			"					4Q		"	"
	I_{IL1}	3009	17		0.4 V			5.5 V							"	4.5 V				1D	-1.4	-3.2	mA
	"	"	18			0.4 V		"							"	4.5 V				2D	"	"	"
	"	"	19				4.5 V	"	0.4 V						"					3D	"	"	"
	"	"	20				4.5 V	"		0.4 V					"					4D	"	"	"
	I_{IL2}	"	21		0.8 V	0.8 V		"							"	0.4 V				CLK 1-2	-2.8	-6.4	"
	I_{IL2}	"	22				0.4 V	"	0.8 V	0.8 V					"					CLK 3-4	-2.8	-6.4	"
	I_{IH1}	3010	23		2.4 V			"							"	GND				1D		80	µA
	"	"	24			2.4 V		"							"	GND				2D		"	"
	"	"	25				GND	"	2.4 V						"					3D		"	"
	"	"	26				GND	"		2.4 V					"					4D		"	"
	I_{IH2}	"	27		5.5 V			"							"	GND				1D		200	"
	"	"	28			5.5 V		"							"	GND				2D		"	"
	"	"	29				GND	"	5.5 V						"					3D		"	"
	"	"	30				GND	"		5.5 V					"					4D		"	"
	I_{IH3}	"	31		GND	GND		"							"	2.4 V				CLK 1-2		160	"
	I_{IH3}	"	32				2.4 V	"	GND	GND					"					CLK 3-4		160	"
	I_{IH4}	"	33		GND	GND		"							"	5.5 V				CLK 1-2		400	"
	I_{IH4}	"	34				5.5 V	"	GND	GND					"					CLK 3-4		400	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 $\overline{Q}$	2 1D	3 2D	4 CLOCK 3-4	5 V_{CC}	6 3D	7 4D	8 4 $\overline{Q}$	9 4Q	10 3Q	11 3 $\overline{Q}$	12 GND	13 CLOCK 1-2	14 2 $\overline{Q}$	15 2Q	16 1Q	Meas. terminal	Test limits Min	Max	Unit
1	I_{OS}	3011	35	GND	GND			5..5 V							GND	4.5 V				1 $\overline{Q}$	-20	-57	mA
T_C = 25°C	"	"	36		4.5 V			"							"	"			GND	1Q	"	"	"
	"	"	37			GND		"							"	"	GND			2 $\overline{Q}$	"	"	"
	"	"	38			4.5 V		"							"	"		GND		2Q	"	"	"
	"	"	39				4.5 V	"	GND					GND	"					3 $\overline{Q}$	"	"	"
	"	"	40				"	"	4.5 V				GND		"					3Q	"	"	"
	"	"	41				"	"		GND	GND				"					4 $\overline{Q}$	"	"	"
	"	"	42				"	"		4.5 V		GND			"					4Q	"	"	"
	I_{CC}	3005	43		GND	GND	GND	"	GND	GND					"	GND				V_{CC}		46	"
	V_{IC}		44		-12 mA			4.5 V							"					1D		-1.5	V
	"		45					"							"	-12 mA				CLK 1-2		"	"
	"		46			-12 mA		"							"					2D		"	"
	"		47					"	-12 mA						"					3D		"	"
	"		48				-12 mA	"							"					CLK 3-4		"	"
	"		49					"		-12 mA					"					4D		"	"
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 1/	Truth	3014	50		B	B	B	5.0 V	B	B					GND	B							
T_C = 25°C	table	"	51		B	B	A	"	B	B					"	A							
	test	"	52		A	A	A	"	A	A					"	A							
	"	"	53	L	A	A	B	"	A	A	L	H	H	L	"	B	L	H	H				
	"	"	54		A	A	B	"	A	A					"	B					2/		
	"	"	55		A	A	A	"	A	A					"	A							
	"	"	56		B	B	A	"	B	B					"	A							
	"	"	57	H	B	B	B	"	B	B	H	L	L	H	"	B	H	L	L				
	"	"	58		B	B	B	"	B	B					"	B							
	"	"	59		A	A	B	"	A	A					"	B							
	"	"	60		A	A	A	"	A	A					"	A							
	"	"	61	L	A	A	B	"	A	A	L	H	H	L	"	B	L	H	H				
	"	"	62		A	A	B	"	A	A					"	B							
	"	"	63		B	B	B	"	B	B					"	B							
	"	"	64		B	B	A	"	B	B					"	A							
	"	"	65	H	B	B	A	"	B	B	H	L	L	H	"	A	H	L	L				
8 1/ 3/	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and T_C = -55°C.																						

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	1 $\overline{Q}$	1D	2D	CLOCK 3-4	V_{CC}	3D	4D	4 $\overline{Q}$	4Q	3Q	3 $\overline{Q}$	GND	CLOCK 1-2	2 $\overline{Q}$	2Q	1Q		Min	Max	Unit
9	t_{PLH1}	3003	66		IN-W			5.0 V							GND	A			OUT	1D-1Q	2	34	ns
T_C = 25°C	"	(Fig 4)	67			IN-W		"							"	A		OUT		2D-2Q	"	"	"
	"	"	68				A	"	IN-W				OUT		"					3D-3Q	"	"	"
	"	"	69				A	"		IN-W		OUT			"					4D-4Q	"	"	"
	t_{PLH2}	"	70	OUT	IN-X			"							"	A				1D-1 $\overline{Q}$	"	44	"
	"	"	71			IN-X		"							"	A	OUT			2D-2 $\overline{Q}$	"	"	"
	"	"	72				A	"	IN-X					OUT	"					3D-3 $\overline{Q}$	"	"	"
	"	"	73				A	"		IN-X	OUT				"					4D-4 $\overline{Q}$	"	"	"
	t_{PLH3}	3003	74		IN			"							"	IN			OUT	CLK 1-2-1Q	"	34	"
	"	(Fig 5)	75			IN		"							"	IN		OUT		CLK 1-2-2Q	"	"	"
	"	"	76				IN	"	IN				OUT		"					CLK 3-4-3Q	"	"	"
	"	"	77				IN	"		IN		OUT			"					CLK 3-4-4Q	"	"	"
	t_{PLH4}	"	78	OUT	IN			"							"	IN				CLK 1-2-1 $\overline{Q}$	"	"	"
	"	"	79			IN		"							"	IN	OUT			CLK 1-2-2 $\overline{Q}$	"	"	"
	"	"	80				IN	"	IN					OUT	"					CLK 3-4-3 $\overline{Q}$	"	"	"
	"	"	81				IN	"		IN	OUT				"					CLK 3-4-4 $\overline{Q}$	"	"	"
	t_{PHL1}	3003	82		IN-X			"							"	A			OUT	1D-1Q	"	29	"
	"	(Fig 4)	83			IN-X		"							"	A		OUT		2D-2Q	"	"	"
	"	"	84				A	"	IN-X				OUT		"					3D-3Q	"	"	"
	"	"	85				A	"		IN-X		OUT			"					4D-4Q	"	"	"
	t_{PHL2}	"	86	OUT	IN-W			"							"	A				1D-1 $\overline{Q}$	"	19	"
	"	"	87			IN-W		"							"	A	OUT			2D-2 $\overline{Q}$	"	"	"
	"	"	88				A	"	IN-W					OUT	"					3D-3 $\overline{Q}$	"	"	"
	"	"	89				A	"		IN-W	OUT				"					4D-4 $\overline{Q}$	"	"	"
	t_{PHL3}	3003	90		IN			"							"	IN			OUT	CLK 1-2-1Q	"	"	"
	"	(Fig 5)	91			IN		"							"	IN		OUT		CLK 1-2-2Q	"	"	"
	"	"	92				IN	"	IN				OUT		"					CLK 3-4-3Q	"	"	"
	"	"	93				IN	"		IN		OUT			"					CLK 3-4-4Q	"	"	"
	t_{PHL4}	"	94	OUT	IN			"							"	IN				CLK 1-2-1 $\overline{Q}$	"	"	"
	"	"	95			IN		"							"	IN	OUT			CLK 1-2-2 $\overline{Q}$	"	"	"
	"	"	96				IN	"	IN					OUT	"					CLK 3-4-3 $\overline{Q}$	"	"	"
	"	"	97				IN	"		IN	OUT				"					CLK 3-4-4 $\overline{Q}$	"	"	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	1 $\overline{Q}$	1D	2D	CLOCK 3-4	V_{CC}	3D	4D	4 $\overline{Q}$	4Q	3Q	3 $\overline{Q}$	GND	CLOCK 1-2	2 $\overline{Q}$	2Q	1Q		Min	Max	Unit
10	t_{PLH1}	3003	98		IN-W			5.0 V							GND	A			OUT	1D-1Q	2	44	ns
T_C = 125°C	"	(Fig 4)	99			IN-W		"							"	A		OUT		2D-2Q	"	"	"
	"	"	100				A	"	IN-W				OUT		"					3D-3Q	"	"	"
	"	"	101				A	"		IN-W		OUT			"					4D-4Q	"	"	"
	t_{PLH2}	"	102	OUT	IN-X			"							"	A				1D-1 $\overline{Q}$	"	55	"
	"	"	103			IN-X		"							"	A	OUT			2D-2 $\overline{Q}$	"	"	"
	"	"	104				A	"	IN-X					OUT	"					3D-3 $\overline{Q}$	"	"	"
	"	"	105				A	"		IN-X	OUT				"					4D-4 $\overline{Q}$	"	"	"
	t_{PLH3}	3003	106		IN			"							"	IN			OUT	CLK 1-2-1Q	"	44	"
	"	(Fig 5)	107			IN		"							"	IN		OUT		CLK 1-2-2Q	"	"	"
	"	"	108				IN	"	IN				OUT		"					CLK 3-4-3Q	"	"	"
	"	"	109				IN	"		IN		OUT			"					CLK 3-4-4Q	"	"	"
	t_{PLH4}	"	110	OUT	IN			"							"	IN				CLK 1-2-1 $\overline{Q}$	"	"	"
	"	"	111			IN		"							"	IN	OUT			CLK 1-2-2 $\overline{Q}$	"	"	"
	"	"	112				IN	"	IN					OUT	"					CLK 3-4-3 $\overline{Q}$	"	"	"
	"	"	113				IN	"		IN	OUT				"					CLK 3-4-4 $\overline{Q}$	"	"	"
	t_{PHL1}	3003	114		IN-X			"							"	A			OUT	1D-1Q	"	38	"
	"	(Fig 4)	115			IN-X		"							"	A		OUT		2D-2Q	"	"	"
	"	"	116				A	"	IN-X				OUT		"					3D-3Q	"	"	"
	"	"	117				A	"		IN-X		OUT			"					4D-4Q	"	"	"
	t_{PHL2}	"	118	OUT	IN-W			"							"	A				1D-1 $\overline{Q}$	"	25	"
	"	"	119			IN-W		"							"	A	OUT			2D-2 $\overline{Q}$	"	"	"
	"	"	120				A	"	IN-W					OUT	"					3D-3 $\overline{Q}$	"	"	"
	"	"	121				A	"		IN-W	OUT				"					4D-4 $\overline{Q}$	"	"	"
	t_{PHL3}	3003	122		IN			"							"	IN			OUT	CLK 1-2-1Q	"	"	"
	"	(Fig 5)	123			IN		"							"	IN		OUT		CLK 1-2-2Q	"	"	"
	"	"	124				IN	"	IN				OUT		"					CLK 3-4-3Q	"	"	"
	"	"	125				IN	"		IN		OUT			"					CLK 3-4-4Q	"	"	"
	t_{PHL4}	"	126	OUT	IN			"							"	IN				CLK 1-2-1 $\overline{Q}$	"	"	"
	"	"	127			IN		"							"	IN	OUT			CLK 1-2-2 $\overline{Q}$	"	"	"
	"	"	128				IN	"	IN					OUT	"					CLK 3-4-3 $\overline{Q}$	"	"	"
	"	"	129				IN	"		IN	OUT				"					CLK 3-4-4 $\overline{Q}$	"	"	"

See footnotes on next page.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

NOTES:

A = Clock pulse A, see figure 4.

W and X = Various D-input pulses, see figure 4.

1/ Only a summary of attributes data is required.

2/ Only voltages shall be either:

(a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or

(b) H ≥ 1.5 volts and L ≤ 1.5 volts when using a high speed checker single comparator.

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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Test No.	1D	2D	CLOCK 3-4	V_{CC}	3D	4D	NC	4Q	3Q	NC	GND	CLOCK 1-2	2Q	1Q		Min	Max	Unit
1	V_{OH}	3006	1	2.0 V			4.5 V							GND	2.0 V		-0.4 mA	1Q	2.4		V
T_C =25°C	"	"	2		2.0 V		"							"	2.0 V	-0.4 mA		2Q	"		"
"	"	"	3			2.0 V	"	2.0 V				-0.4 mA		"				3Q	"		"
"	"	"	4			2.0 V	"		2.0 V		-0.4 mA			"				4Q	"		"
	V_{OL}	3007	5	0.8 V			"							"	2.0 V		16 mA	1Q		0.4	"
	"	"	6		0.8 V		"							"	2.0 V	16 mA		2Q		"	"
	"	"	7			2.0 V	"	0.8 V				16 mA		"				3Q		"	"
	"	"	8			2.0 V	"		0.8 V		16 mA			"				4Q		"	"
	I_{IL1}	3009	9	0.4 V			5.5 V							"	4.5 V			1D	-1.4	-3.2	mA
	"	"	10		0.4 V		"							"	4.5 V			2D	"	"	"
	"	"	11			4.5 V	"	0.4 V						"				3D	"	"	"
	"	"	12			4.5 V	"		0.4 V					"				4D	"	"	"
	I_{IL2}	"	13	0.8 V	0.8 V		"							"	0.4 V			CLK 1-2	-2.8	-6.4	"
	I_{IL2}	"	14			0.4 V	"	0.8 V	0.8 V					"				CLK 3-4	-2.8	-6.4	"
	I_{IH1}	3010	15	2.4 V			"							"	GND			1D		80	µA
	"	"	16		2.4 V		"							"	GND			2D		"	"
	"	"	17			GND	"	2.4 V						"				3D		"	"
	"	"	18			GND	"		2.4 V					"				4D		"	"
	I_{IH2}	"	19	5.5 V			"							"	GND			1D		200	"
	"	"	20		5.5 V		"							"	GND			2D		"	"
	"	"	21			GND	"	5.5 V						"				3D		"	"
	"	"	22			GND	"		5.5 V					"				4D		"	"
	I_{IH3}	"	23	GND	GND		"							"	2.4 V			CLK 1-2		160	"
	I_{IH3}	"	24			2.4 V	"	GND	GND					"				CLK 3-4		160	"
	I_{IH4}	"	25	GND	GND		"							"	5.5 V			CLK 1-2		400	"
	I_{IH4}	"	26			5.5 V	"	GND	GND					"				CLK 3-4		400	"
	I_{OS}	3011	27	4.5 V			"							"	4.5 V		GND	1Q	-20	-57	mA
	"	"	28		4.5 V		"							"	4.5 V	GND		2Q	"	"	"
	"	"	29			4.5 V	"	4.5 V				GND		"				3Q	"	"	"
	"	"	30			4.5 V	"		4.5 V		GND			"				4Q	"	"	"
	I_{CC}	3005	31	GND	GND	GND	"	GND	GND					"	GND			V_{CC}		46	"
	V_{IC}		32	-12 mA			4.5 V							"				1D		-1.5	V
	"		33		-12 mA		"							"				2D		"	"
	"		34				"	-12 mA						"				3D		"	"
	"		35				"		-12 mA					"				4D		"	"
	"		36				"							"	-12 mA			CLK 1-2		"	"
	"		37			-12 mA	"							"				CLK 3-4		"	"
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Test No.	1D	2D	CLOCK 3-4	V_{CC}	3D	4D	NC	4Q	3Q	NC	GND	CLOCK 1-2	2Q	1Q		Min	Max	Unit
7 1/	VLT-H1	see fig 4	38	IN-W			5.0 V							GND	A		OUT	1D-1Q	2.4		V
T_C =25°C	"		39		IN-W		"							"	A	OUT		2D-2Q	"		"
	"		40			A	"	IN-W				OUT		"				3D-3Q	"		"
	"		41			A	"		IN-W		OUT			"				4D-4Q	"		"
	VLT-L1		42	IN-X			"							"	A		OUT	1D-1Q		0.4	"
	"		43		IN-X		"							"	A	OUT		2D-2Q		"	"
	"		44			A	"	IN-X				OUT		"				3D-3Q		"	"
	"		45			A	"		IN-X		OUT			"				4D-4Q		"	"
	VLT-H3	see fig 5	46	IN			"							"	IN-B		OUT	CLK 1-2-1Q	2.4		"
	"		47		IN		"							"	IN-B	OUT		CLK 1-2-2Q	"		"
	"		48			IN	"	IN				OUT		"				CLK 3-4-3Q	"		"
	"		49			IN	"		IN		OUT			"				CLK 3-4-4Q	"		"
	VLT-L3		50	IN			"							"	IN-B		OUT	CLK 1-2-1Q		0.4	"
	"		51		IN		"							"	IN-B	OUT		CLK 1-2-2Q		"	"
	"		52			IN	"	IN				OUT		"				CLK 3-4-3Q		"	"
	"		53			IN	"		IN		OUT			"				CLK 3-4-4Q		"	"
8 1/	Same tests, terminal conditions and limits as for subgroup 7, except T_C = 125°C and -55°C.																				
9	t_{PLH1}	3003	54	IN-W			5.0 V							GND	A		OUT	1D-1Q	2	34	ns
T_C =25°C	"	Fig 4-A	55		IN-W		"							"	A	OUT		2D-2Q	"	"	"
	"	"	56			A	"	IN-W				OUT		"				3D-3Q	"	"	"
	"	"	57			A	"		IN-W		OUT			"				4D-4Q	"	"	"
	t_{PHL1}	3003	58	IN-X			"							"	A		OUT	1D-1Q	"	29	"
	"	Fig 4-B	59		IN-X		"							"	A	OUT		2D-2Q	"	"	"
	"	"	60			A	"	IN-X				OUT		"				3D-3Q	"	"	"
	"	"	61			A	"		IN-X		OUT			"				4D-4Q	"	"	"
	t_{PLH3}	3003	62	IN			"							"	IN		OUT	CLK 1-2-1Q	"	34	"
	"	Fig 5	63		IN		"							"	IN	OUT		CLK 1-2-2Q	"	"	"
	"	"	64			IN	"	IN				OUT		"				CLK 3-4-3Q	"	"	"
	"	"	65			IN	"		IN		OUT			"				CLK 3-4-4Q	"	"	"
	t_{PHL3}	"	66	IN			"							"	IN		OUT	CLK 1-2-1Q	"	19	"
	"	"	67		IN		"							"	IN	OUT		CLK 1-2-2Q	"	"	"
	"	"	68			IN	"	IN				OUT		"				CLK 3-4-3Q	"	"	"
	"	"	69			IN	"		IN		OUT			"				CLK 3-4-4Q	"	"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Test No.	1D	2D	CLOCK 3-4	V_{CC}	3D	4D	NC	4Q	3Q	NC	GND	CLOCK 1-2	2Q	1Q		Min	Max	Unit
10	t_{PLH1}	3003	70	IN-W			5.0 V							GND	A		OUT	1D-1Q	2	44	ns
T_C = 125°C	"	Fig 4-A	71		IN-W		"							"	A	OUT		2D-2Q	"	"	"
	"	"	72			A	"	IN-W				OUT		"				3D-3Q	"	"	"
	"	"	73			A	"		IN-W		OUT			"				4D-4Q	"	"	"
	t_{PHL1}	3003	74	IN-X			"							"	A		OUT	1D-1Q	"	38	"
	"	Fig 4-B	75		IN-X		"							"	A	OUT		2D-2Q	"	"	"
	"	"	76			A	"	IN-X				OUT		"				3D-3Q	"	"	"
	"	"	77			A	"		IN-X		OUT			"				4D-4Q	"	"	"
	t_{PLH3}	3003	78	IN			"							"	IN		OUT	CLK 1-2-1Q	"	44	"
	"	Fig 5	79		IN		"							"	IN	OUT		CLK 1-2-2Q	"	"	"
	"	"	80			IN	"	IN				OUT		"				CLK 3-4-3Q	"	"	"
	"	"	81			IN	"		IN		OUT			"				CLK 3-4-4Q	"	"	"
	t_{PHL3}	"	82	IN			"							"	IN		OUT	CLK 1-2-1Q	"	25	"
	"	"	83		IN		"							"	IN	OUT		CLK 1-2-2Q	"	"	"
	"	"	84			IN	"	IN				OUT		"				CLK 3-4-3Q	"	"	"
	"	"	85			IN	"		IN		OUT			"				CLK 3-4-4Q	"	"	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

NOTES:
A = Clock pulse A, see figure 4.
W and X = Various D-input pulses, see figure 4.

1/ Latch voltage, V_{LT}, shall be measured no sooner than 10 μs after switching transition occurs.

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	1	2	3	4	5	6	7	8	9	10	11	12	Meas. terminal	Test limits		
			Test No.	1 $\overline{MR}$	1 $\overline{E}$ 0	1 $\overline{E}$ 1	1D0	1Q0	1D1	1Q1	1D2	1Q2	1D3	1Q3	GND		Min	Max	Unit
1	V_{OH}	3006	1	4.5 V	0.8 V	0.8 V	4.5 V	-0.8 mA	4.5 V		4.5 V		4.5 V		GND	1Q0	2.4		V
T_C = 25°C	"	"	2	"	"	"	"		"	-0.8 mA	"		"		"	1Q1	"		"
"	"	"	3	"	"	"	"		"		"	-0.8 mA	"		"	1Q2	"		"
"	"	"	4	"	"	"	"		"		"		"	-0.8 mA	"	1Q3	"		"
"	"	"	5	"	"	"	"		"		"		"		"	2Q0	"		"
"	"	"	6	"	"	"	"		"		"		"		"	2Q1	"		"
"	"	"	7	"	"	"	"		"		"		"		"	2Q2	"		"
"	"	"	8	"	"	"	"		"		"		"		"	2Q3	"		"
"	V_{OL}	3007	9	"	"	"	0.8 V	16 mA	0.8 V		0.8 V		0.8 V		"	1Q0		0.4	"
"	"	"	10	"	"	"	"		"	16 mA	"		"		"	1Q1		"	"
"	"	"	11	"	"	"	"		"		"	16 mA	"		"	1Q2		"	"
"	"	"	12	"	"	"	"		"		"		"	16 mA	"	1Q3		"	"
"	"	"	13	"	"	"	"		"		"		"		"	2Q0		"	"
"	"	"	14	"	"	"	"		"		"		"		"	2Q1		"	"
"	"	"	15	"	"	"	"		"		"		"		"	2Q2		"	"
"	"	"	16	"	"	"	"		"		"		"		"	2Q3		"	"
"	V_{IC}		17	-12 mA											"	1 $\overline{MR}$		-1.5	"
"	"		18		-12 mA										"	1 $\overline{E}$ 0		"	"
"	"		19			-12 mA									"	1 $\overline{E}$ 1		"	"
"	"		20				-12 mA								"	1D0		"	"
"	"		21						-12 mA						"	1D1		"	"
"	"		22								-12 mA				"	1D2		"	"
"	"		23										-12 mA		"	1D3		"	"
"	"		24												"	2 $\overline{MR}$		"	"
"	"		25												"	2 $\overline{E}$ 0		"	"
"	"		26												"	2 $\overline{E}$ 1		"	"
"	"		27												"	2D0		"	"
"	"		28												"	2D1		"	"
"	"		29												"	2D2		"	"
"	"		30												"	2D3		"	"
"	I_{IL1}	3009	31 CKT A	GND	GND		0.4 V								"	1D0	-0.7	-2.4	mA
"	"	"	31 CKT B	"	"		"								"	1D0	-0.7	-1.6	"
"	"	"	31 CKT C	"	"		"								"	1D0	-0.5	-1.4	"
"	"	"	32 CKT A	"	"				0.4 V						"	1D1	-0.7	-2.4	"
"	"	"	32 CKT B	"	"				"						"	1D1	-0.7	-1.6	"
"	"	"	32 CKT C	"	"				"						"	1D1	-0.5	-1.4	"
"	"	"	33 CKT A	"	"						0.4 V				"	1D2	-0.7	-2.4	"
"	"	"	33 CKT B	"	"						"				"	1D2	-0.7	-1.6	"
"	"	"	33 CKT C	"	"						"				"	1D2	-0.5	-1.4	"
"	"	"	34 CKT A	"	"								0.4 V		"	1D3	-0.7	-2.4	"
"	"	"	34 CKT B	"	"								"		"	1D3	-0.7	-1.6	"
"	"	"	34 CKT C	"	"								"		"	1D3	-0.5	-1.4	"
"	"	"	35 CKT A												"	2D0	-0.7	-2.4	"
"	"	"	35 CKT B												"	2D0	-0.7	-1.6	"
"	"	"	35 CKT C												"	2D0	-0.5	-1.4	"
"	"	"	36 CKT A												"	2D1	-0.7	-2.4	"
"	"	"	36 CKT B												"	2D1	-0.7	-1.6	"
"	"	"	36 CKT C												"	2D1	-0.5	-1.4	"

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	13	14	15	16	17	18	19	20	21	22	23	24	Meas. terminal	Test limits		
			Test No.	2 $\overline{MR}$	2 $\overline{E}$ 0	2 $\overline{E}$ 1	2D0	2Q0	2D1	2Q1	2D2	2Q2	2D3	2Q3	V_{CC}		Min	Max	Unit
1	V_{OH}	3006	1	4.5 V	0.8 V	0.8 V	4.5 V		4.5 V		4.5 V		4.5 V		GND	1Q0	2.4		V
T_C = 25°C	"	"	2	"	"	"	"		"		"		"		"	1Q1	"		"
"	"	"	3	"	"	"	"		"		"		"		"	1Q2	"		"
"	"	"	4	"	"	"	"		"		"		"		"	1Q3	"		"
"	"	"	5	"	"	"	"	-0.8 mA	"		"		"		"	2Q0	"		"
"	"	"	6	"	"	"	"		"	-0.8 mA	"		"		"	2Q1	"		"
"	"	"	7	"	"	"	"		"		"	-0.8 mA	"		"	2Q2	"		"
"	"	"	8	"	"	"	"		"		"		"	-0.8 mA	"	2Q3	"		"
"	V_{OL}	3007	9	"	"	"	0.8 V		0.8 V		0.8 V		0.8 V		"	1Q0		0.4	"
"	"	"	10	"	"	"	"		"		"		"		"	1Q1		"	"
"	"	"	11	"	"	"	"		"		"		"		"	1Q2		"	"
"	"	"	12	"	"	"	"		"		"		"		"	1Q3		"	"
"	"	"	13	"	"	"	"	16 mA	"		"		"		"	2Q0		"	"
"	"	"	14	"	"	"	"		"	16 mA	"		"		"	2Q1		"	"
"	"	"	15	"	"	"	"		"		"	16 mA	"		"	2Q2		"	"
"	"	"	16	"	"	"	"		"		"		"	16 mA	"	2Q3		"	"
"	V_{IC}		17												"	1 $\overline{MR}$		-1.5	"
"	"		18												"	1 $\overline{E}$ 0		"	"
"	"		19												"	1 $\overline{E}$ 1		"	"
"	"		20												"	1D0		"	"
"	"		21												"	1D1		"	"
"	"		22												"	1D2		"	"
"	"		23												"	1D3		"	"
"	"		24	-12 mA											"	2 $\overline{MR}$		"	"
"	"		25		-12 mA										"	2 $\overline{E}$ 0		"	"
"	"		26			-12 mA									"	2 $\overline{E}$ 1		"	"
"	"		27				-12 mA								"	2D0		"	"
"	"		28						-12 mA						"	2D1		"	"
"	"		29								-12 mA				"	2D2		"	"
"	"		30										-12 mA		"	2D3		"	"
"	I_{IL1}	3009	31 CKT A												"	1D0	-0.7	-2.4	mA
"	"	"	31 CKT B												"	1D0	-0.7	-1.6	"
"	"	"	31 CKT C												"	1D0	-0.5	-1.4	"
"	"	"	32 CKT A												"	1D1	-0.7	-2.4	"
"	"	"	32 CKT B												"	1D1	-0.7	-1.6	"
"	"	"	32 CKT C												"	1D1	-0.5	-1.4	"
"	"	"	33 CKT A												"	1D2	-0.7	-2.4	"
"	"	"	33 CKT B												"	1D2	-0.7	-1.6	"
"	"	"	33 CKT C												"	1D2	-0.5	-1.4	"
"	"	"	34 CKT A												"	1D3	-0.7	-2.4	"
"	"	"	34 CKT B												"	1D3	-0.7	-1.6	"
"	"	"	34 CKT C												"	1D3	-0.5	-1.4	"
"	"	"	35 CKT A	GND	GND		-0.4 V								"	2D0	-0.7	-2.4	"
"	"	"	35 CKT B	"	"		"								"	2D0	-0.7	-1.6	"
"	"	"	35 CKT C	"	"		"								"	2D0	-0.5	-1.4	"
"	"	"	36 CKT A	"	"				-0.4 V						"	2D1	-0.7	-2.4	"
"	"	"	36 CKT B	"	"				"						"	2D1	-0.7	-1.6	"
"	"	"	36 CKT C	"	"				"						"	2D1	-0.5	-1.4	"

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	1	2	3	4	5	6	7	8	9	10	11	12	Meas. terminal	Test limits		
			Test No.	1 $\overline{MR}$	1 $\overline{E}$ 0	1 $\overline{E}$ 1	1D0	1Q0	1D1	1Q1	1D2	1Q2	1D3	1Q3	GND		Min	Max	Unit
1	I_{IL1}	3009	37 CKT A												GND	2D2	-0.7	-2.4	mA
T_C = 25°C	"	"	37 CKT B												"	2D2	-0.7	-1.6	"
"	"	"	37 CKT C												"	2D2	-0.5	-1.4	"
"	"	"	38 CKT A												"	2D3	-0.7	-2.4	"
"	"	"	38 CKT B												"	2D3	-0.7	-1.6	"
"	"	"	38 CKT C												"	2D3	-0.5	-1.4	"
"	I_{IL2}	"	39	0.4 V											"	1 $\overline{MR}$	-0.7	-1.6	"
""	"	"	40		0.4 V	5.5 V									"	1 $\overline{E}$ 0	"	"	"
"	"	"	41		5.5 V	0.4 V									"	1 $\overline{E}$ 1	"	"	"
"	"	"	42												"	2 $\overline{MR}$	"	"	"
"	"	"	43												"	2 $\overline{E}$ 0	"	"	"
"	"	"	44												"	2 $\overline{E}$ 1	"	"	"
"	I_{IH1}	3010	45 CKT A			5.5 V	2.4 V								"	1D0		40	µA
"	"	"	45 CKT B, C			"	2.4 V								"	1D0		80	"
"	"	"	46 CKT A			"			2.4 V						"	1D1		40	"
"	"	"	46 CKT B, C			"			2.4 V						"	1D1		80	"
"	"	"	47 CKT A			"					2.4 V				"	1D2		40	"
"	"	"	47 CKT B, C			"					2.4 V				"	1D2		80	"
"	"	"	48 CKT A			"							2.4 V		"	1D3		40	"
"	"	"	48 CKT B, C			"							2.4 V		"	1D3		80	"
"	"	"	49 CKT A												"	2D0		40	"
"	"	"	49 CKT B, C												"	2D0		80	"
"	"	"	50 CKT A												"	2D1		40	"
"	"	"	50 CKT B, C												"	2D1		80	"
"	"	"	51 CKT A												"	2D2		40	"
"	"	"	51 CKT B, C												"	2D2		80	"
"	"	"	52 CKT A												"	2D3		40	"
"	"	"	52 CKT B, C												"	2D3		80	"
"	I_{IH2}	3010	53 CKT A			5.5 V	5.5 V								"	1D0		100	"
"	"	"	53 CKT B, C			"	5.5 V								"	1D0		200	"
"	"	"	54 CKT A			"			5.5 V						"	1D1		100	"
"	"	"	54 CKT B, C			"			5.5 V						"	1D1		200	"
"	"	"	55 CKT A			"					5.5 V				"	1D2		100	"
"	"	"	55 CKT B, C			"					5.5 V				"	1D2		200	"
"	"	"	56 CKT A			"							5.5 V		"	1D3		100	"
"	"	"	56 CKT B, C			"							5.5 V		"	1D3		200	"
"	"	"	57 CKT A												"	2D0		100	"
"	"	"	57 CKT B, C												"	2D0		200	"
"	"	"	58 CKT A												"	2D1		100	"
"	"	"	58 CKT B, C												"	2D1		200	"
"	"	"	59 CKT A												"	2D2		100	"
"	"	"	59 CKT B, C												"	2D2		200	"
"	"	"	60 CKT A												"	2D3		100	"
"	"	"	60 CKT B, C												"	2D3		200	"

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	13	14	15	16	17	18	19	20	21	22	23	24	Meas. terminal	Test limits		
			Test No.	$2\overline{MR}$	$2\overline{E}\ 0$	$2\overline{E}\ 1$	2D0	2Q0	2D1	2Q1	2D2	2Q2	2D3	2Q3	V_{CC}		Min	Max	Unit
1	I_{IL1}	3009	37 CKT A	GND	GND						0.4 V				5.5 V	2D2	-0.7	-2.4	mA
T_C = 25°C	"	"	37 CKT B	"	"						"				"	2D2	-0.7	-1.6	"
"	"	"	37 CKT C	"	"						"				"	2D2	-0.5	-1.4	"
"	"	"	38 CKT A	"	"								0.4 V		"	2D3	-0.7	-2.4	"
"	"	"	38 CKT B	"	"								"		"	2D3	-0.7	-1.6	"
"	"	"	38 CKT C	"	"								"		"	2D3	-0.5	-1.4	"
"	I_{IL2}	"	39												"	$1\overline{MR}$	-0.7	-1.6	"
""	"	"	40												"	$1\overline{E}\ 0$	"	"	"
"	"	"	41												"	$1\overline{E}\ 1$	"	"	"
"	"	"	42	0.4 V											"	$2\overline{MR}$	"	"	"
"	"	"	43		0.4 V	5.5 V									"	$2\overline{E}\ 0$	"	"	"
"	"	"	44		5.5 V	0.4 V									"	$2\overline{E}\ 1$	"	"	"
"	I_{IH1}	3010	45 CKT A												"	1D0		40	µA
"	"	"	45 CKT B, C												"	1D0		80	"
"	"	"	46 CKT A												"	1D1		40	"
"	"	"	46 CKT B, C												"	1D1		80	"
"	"	"	47 CKT A												"	1D2		40	"
"	"	"	47 CKT B, C												"	1D2		80	"
"	"	"	48 CKT A												"	1D3		40	"
"	"	"	48 CKT B, C												"	1D3		80	"
"	"	"	49 CKT A			5.5 V	2.4 V								"	2D0		40	"
"	"	"	49 CKT B, C			"	2.4 V								"	2D0		80	"
"	"	"	50 CKT A			"			2.4 V						"	2D1		40	"
"	"	"	50 CKT B, C			"			2.4 V						"	2D1		80	"
"	"	"	51 CKT A			"					2.4 V				"	2D2		40	"
"	"	"	51 CKT B, C			"					2.4 V				"	2D2		80	"
"	"	"	52 CKT A			"							2.4 V		"	2D3		40	"
"	"	"	52 CKT B, C			"							2.4 V		"	2D3		80	"
"	I_{IH2}	3010	53 CKT A												"	1D0		100	"
"	"	"	53 CKT B, C												"	1D0		200	"
"	"	"	54 CKT A												"	1D1		100	"
"	"	"	54 CKT B, C												"	1D1		200	"
"	"	"	55 CKT A												"	1D2		100	"
"	"	"	55 CKT B, C												"	1D2		200	"
"	"	"	56 CKT A												"	1D3		100	"
"	"	"	56 CKT B, C												"	1D3		200	"
"	"	"	57 CKT A			5.5 V	5.5 V								"	2D0		100	"
"	"	"	57 CKT B, C			"	5.5 V								"	2D0		200	"
"	"	"	58 CKT A			"			5.5 V						"	2D1		100	"
"	"	"	58 CKT B, C			"			5.5 V						"	2D1		200	"
"	"	"	59 CKT A			"					5.5 V				"	2D2		100	"
"	"	"	59 CKT B, C			"					5.5 V				"	2D2		200	"
"	"	"	60 CKT A			"							5.5 V		"	2D3		100	"
"	"	"	60 CKT B, C			"							5.5 V		"	2D3		200	"

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	1	2	3	4	5	6	7	8	9	10	11	12	Meas. terminal	Test limits		
			Test No.	1 $\overline{MR}$	1 $\overline{E}$ 0	1 $\overline{E}$ 1	1D0	1Q0	1D1	1Q1	1D2	1Q2	1D3	1Q3	GND		Min	Max	Unit
1	I_{IH3}	3010	61	2.4 V											GND	1 $\overline{MR}$		40	µA
T_C = 25°C	"	"	62		2.4 V	GND									"	1 $\overline{E}$ 0		"	"
"	"	"	63		GND	2.4 V									"	1 $\overline{E}$ 1		"	"
"	"	"	64												"	2 $\overline{MR}$		"	"
"	"	"	65												"	2 $\overline{E}$ 0		"	"
"	"	"	66												"	2 $\overline{E}$ 1		"	"
"	I_{IH4}	"	67	5.5 V											"	1 $\overline{MR}$		100	"
"	"	"	68		5.5 V	GND									"	1 $\overline{E}$ 0		"	"
"	"	"	69		GND	5.5 V									"	1 $\overline{E}$ 1		"	"
"	"	"	70												"	2 $\overline{MR}$		"	"
"	"	"	71												"	2 $\overline{E}$ 0		"	"
"	"	"	72												"	2 $\overline{E}$ 1		"	"
"	I_{OS}	3011	73	4.5 V	GND	GND	4.5 V	GND	4.5 V		4.5 V		4.5 V		"	1Q0	-10 1/	-70	mA
"	"	"	74	"	"	"	"		"	GND	"		"		"	1Q1	"	"	"
"	"	"	75	"	"	"	"		"		"	GND	"		"	1Q2	"	"	"
"	"	"	76	"	"	"	"		"		"		"	GND	"	1Q3	"	"	"
"	"	"	77	"	"	"	"		"		"		"		"	2Q0	"	"	"
"	"	"	78	"	"	"	"		"		"		"		"	2Q1	"	"	"
"	"	"	79	"	"	"	"		"		"		"		"	2Q2	"	"	"
"	"	"	80	"	"	"	"		"		"		"		"	2Q3	"	"	"
"	I_{CC}	3005	81	4.5 V	4.5 V	4.5 V	4.5 V		4.5 V		4.5 V		4.5 V		"	V_{CC}		106	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																		
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																		
9	t_{PLH1}	3003	82	5.0 V	IN	GND	IN	OUT							GND	1 $\overline{E}$ 0-1Q0	3	33	ns
T_C = 25°C	"	Fig 6-B	83	"	"	"			IN	OUT					"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	84	"	"	"					IN	OUT			"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	85	"	"	"							IN	OUT	"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	86	"	GND	IN	IN	OUT							"	1 $\overline{E}$ 0-1Q0	"	"	"
"	"	"	87	"	"	"			IN	OUT					"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	88	"	"	"					IN	OUT			"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	89	"	"	"							IN	OUT	"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	90	"											"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	91	"											"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	92	"											"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	93	"											"	2 $\overline{E}$ 3-2Q3	"	"	"
"	"	"	94	"											"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	95	"											"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	96	"											"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	97	"											"	2 $\overline{E}$ 3-2Q3	"	"	"

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	1	2	3	4	5	6	7	8	9	10	11	12	Meas. terminal	Test limits		
			Test No.	2 $\overline{MR}$	2 $\overline{E}$ 0	2 $\overline{E}$ 1	2D0	2Q0	2D1	2Q1	2D2	2Q2	2D3	2Q3	V_{CC}		Min	Max	Unit
1	I_{IH3}	3010	61												5.5 V	1 $\overline{MR}$		40	µA
T_C = 25°C	"	"	62												"	1 $\overline{E}$ 0		"	"
"	"	"	63												"	1 $\overline{E}$ 1		"	"
"	"	"	64	2.4 V											"	2 $\overline{MR}$		"	"
"	"	"	65		2.4 V	GND									"	2 $\overline{E}$ 0		"	"
"	"	"	66		GND	2.4 V									"	2 $\overline{E}$ 1		"	"
"	I_{IH4}	"	67												"	1 $\overline{MR}$		100	"
"	"	"	68												"	1 $\overline{E}$ 0		"	"
"	"	"	69												"	1 $\overline{E}$ 1		"	"
"	"	"	70	5.5 V											"	2 $\overline{MR}$		"	"
"	"	"	71		5.5 V	GND									"	2 $\overline{E}$ 0		"	"
"	"	"	72		GND	5.5 V									"	2 $\overline{E}$ 1		"	"
"	I_{OS}	3011	73	4.5 V	GND	GND	4.5 V		4.5 V		4.5 V		4.5 V		"	1Q0	-10 1/	-70	mA
"	"	"	74	"	"	"	"		"		"		"		"	1Q1	"	"	"
"	"	"	75	"	"	"	"		"		"		"		"	1Q2	"	"	"
"	"	"	76	"	"	"	"		"		"		"		"	1Q3	"	"	"
"	"	"	77	"	"	"	"	GND	"		"		"		"	2Q0	"	"	"
"	"	"	78	"	"	"	"		"	GND	"		"		"	2Q1	"	"	"
"	"	"	79	"	"	"	"		"		"	GND	"		"	2Q2	"	"	"
"	"	"	80	"	"	"	"		"		"		"	GND	"	2Q3	"	"	"
"	I_{CC}	3005	81	4.5 V	4.5 V	4.5 V	4.5 V		4.5 V		4.5 V		4.5 V		"	V_{CC}		106	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																		
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																		
9	t_{PLH1}	3003	82	5.0 V											5.0 V	1 $\overline{E}$ 0-1Q0	3	33	ns
T_C = 25°C	"	Fig 6-B	83	"											"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	84	"											"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	85	"											"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	86	"											"	1 $\overline{E}$ 0-1Q0	"	"	"
"	"	"	87	"											"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	88	"											"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	89	"											"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	90	"	IN	GND	IN	OUT							"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	91	"	"	"			IN	OUT					"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	92	"	"	"					IN	OUT			"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	93	"	"	"							IN	OUT	"	2 $\overline{E}$ 3-2Q3	"	"	"
"	"	"	94	"	GND	IN	IN	OUT							"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	95	"	"	"			IN	OUT					"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	96	"	"	"					IN	OUT			"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	97	"	"	"							IN	OUT	"	2 $\overline{E}$ 3-2Q3	"	"	"

1/ Circuit B and C limits for I_{OS} shall be -20, minimum and -57, maximum.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	1	2	3	4	5	6	7	8	9	10	11	12	Meas. terminal	Test limits		
			Test No.	1 $\overline{MR}$	1 $\overline{E}$ 0	1 $\overline{E}$ 1	1D0	1Q0	1D1	1Q1	1D2	1Q2	1D3	1Q3	GND		Min	Max	Unit
9	t_{PHL1}	3003	98	5.0 V	IN	GND	IN	OUT							GND	1 $\overline{E}$ 0-1Q0	3	25	ns
T_C = 25°C	"	Fig 6-A	99	"	"	"			IN	OUT					"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	100	"	"	"					IN	OUT			"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	101	"	"	"							IN	OUT	"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	102	"	GND	IN	IN	OUT							"	1 $\overline{E}$ 0-1Q0	"	"	"
"	"	"	103	"	"	"			IN	OUT					"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	104	"	"	"					IN	OUT			"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	105	"	"	"							IN	OUT	"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	106	"											"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	107	"											"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	108	"											"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	109	"											"	2 $\overline{E}$ 3-2Q3	"	"	"
"	"	"	110	"											"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	111	"											"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	112	"											"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	113	"											"	2 $\overline{E}$ 3-2Q3	"	"	"
"	t_{PLH2}	3003	114	"	IN	GND	IN	OUT							"	1D0-1Q0	"	33	"
"	"	Fig 6	115	"	"	"			IN	OUT					"	1D1-1Q1	"	"	"
"	"	"	116	"	"	"					IN	OUT			"	1D2-1Q2	"	"	"
"	"	"	117	"	"	"							IN	OUT	"	1D3-1Q3	"	"	"
"	"	"	118	"	GND	IN	IN	OUT							"	1D0-1Q0	"	"	"
"	"	"	119	"	"	"			IN	OUT					"	1D1-1Q1	"	"	"
"	"	"	120	"	"	"					IN	OUT			"	1D2-1Q2	"	"	"
"	"	"	121	"	"	"							IN	OUT	"	1D3-1Q3	"	"	"
"	"	"	122	"											"	2D0-2Q0	"	"	"
"	"	"	123	"											"	2D1-2Q1	"	"	"
"	"	"	124	"											"	2D2-2Q2	"	"	"
"	"	"	125	"											"	2D3-2Q3	"	"	"
"	"	"	126	"											"	2D0-2Q0	"	"	"
"	"	"	127	"											"	2D1-2Q1	"	"	"
"	"	"	128	"											"	2D2-2Q2	"	"	"
"	"	"	129	"											"	2D3-2Q3	"	"	"

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	13	14	15	16	17	18	19	20	21	22	23	24	Meas. terminal	Test limits		
			Test No.	2 $\overline{MR}$	2 $\overline{E}$ 0	2 $\overline{E}$ 1	2D0	2Q0	2D1	2Q1	2D2	2Q2	2D3	2Q3	V_{CC}		Min	Max	Unit
9	t_{PHL1}	3003	98	5.0 V											5.0 V	1 $\overline{E}$ 0-1Q0	3	25	ns
T_C = 25°C	"	Fig 6-A	99	"											"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	100	"											"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	101	"											"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	102	"											"	1 $\overline{E}$ 0-1Q0	"	"	"
"	"	"	103	"											"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	104	"											"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	105	"											"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	106	"	IN	GND	IN	OUT							"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	107	"	"	"			IN	OUT					"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	108	"	"	"					IN	OUT			"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	109	"	"	"							IN	OUT	"	2 $\overline{E}$ 3-2Q3	"	"	"
"	"	"	110	"	GND	IN	IN	OUT							"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	111	"	"	"			IN	OUT					"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	112	"	"	"					IN	OUT			"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	113	"	"	"							IN	OUT	"	2 $\overline{E}$ 3-2Q3	"	"	"
"	t_{PLH2}	3003	114	"											"	1D0-1Q0	"	33	"
"	"	Fig 6	115	"											"	1D1-1Q1	"	"	"
"	"	"	116	"											"	1D2-1Q2	"	"	"
"	"	"	117	"											"	1D3-1Q3	"	"	"
"	"	"	118	"											"	1D0-1Q0	"	"	"
"	"	"	119	"											"	1D1-1Q1	"	"	"
"	"	"	120	"											"	1D2-1Q2	"	"	"
"	"	"	121	"											"	1D3-1Q3	"	"	"
"	"	"	122	"	IN	GND	IN	OUT							"	2D0-2Q0	"	"	"
"	"	"	123	"	"	"			IN	OUT					"	2D1-2Q1	"	"	"
"	"	"	124	"	"	"					IN	OUT			"	2D2-2Q2	"	"	"
"	"	"	125	"	"	"							IN	OUT	"	2D3-2Q3	"	"	"
"	"	"	126	"	GND	IN	IN	OUT							"	2D0-2Q0	"	"	"
"	"	"	127	"	"	"			IN	OUT					"	2D1-2Q1	"	"	"
"	"	"	128	"	"	"					IN	OUT			"	2D2-2Q2	"	"	"
"	"	"	129	"	"	"							IN	OUT	"	2D3-2Q3	"	"	"

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	1	2	3	4	5	6	7	8	9	10	11	12	Meas. terminal	Test limits		
			Test No.	1 $\overline{MR}$	1 $\overline{E}$ 0	1 $\overline{E}$ 1	1D0	1Q0	1D1	1Q1	1D2	1Q2	1D3	1Q3	GND		Min	Max	Unit
9	t_{PHL2}	3003	130	5.0 V	IN	GND	IN	OUT							GND	1D0-1Q0	3	28	ns
T_C = 25°C	"	Fig 6	131	"	"	"			IN	OUT					"	1D1-1Q1	"	"	"
"	"	"	132	"	"	"					IN	OUT			"	1D2-1Q2	"	"	"
"	"	"	133	"	"	"							IN	OUT	"	1D3-1Q3	"	"	"
"	"	"	134	"	GND	IN	IN	OUT							"	1D0-1Q0	"	"	"
"	"	"	135	"	"	"			IN	OUT					"	1D1-1Q1	"	"	"
"	"	"	136	"	"	"					IN	OUT			"	1D2-1Q2	"	"	"
"	"	"	137	"	"	"							IN	OUT	"	1D3-1Q3	"	"	"
"	"	"	138	"											"	2D0-2Q0	"	"	"
"	"	"	139	"											"	2D1-2Q1	"	"	"
"	"	"	140	"											"	2D2-2Q2	"	"	"
"	"	"	141	"											"	2D3-2Q3	"	"	"
"	"	"	142	"											"	2D0-2Q0	"	"	"
"	"	"	143	"											"	2D1-2Q1	"	"	"
"	"	"	144	"											"	2D2-2Q2	"	"	"
"	"	"	145	"											"	2D3-2Q3	"	"	"
"	t_{PHL3}	3003	146	IN	IN	GND	5.0 V	OUT	5.0 V		5.0 V		5.0 V		"	1 $\overline{MR}$ -1Q0	"	25	"
"	"	Fig 6-C	147	"	"	"	"		"	OUT	"		"		"	1 $\overline{MR}$ -1Q1	"	"	"
"	"	"	148	"	"	"	"		"		"	OUT	"		"	1 $\overline{MR}$ -1Q2	"	"	"
"	"	"	149	"	"	"	"		"		"		"	OUT	"	1 $\overline{MR}$ -1Q3	"	"	"
"	"	"	150	"	GND	IN	"	OUT	"		"		"		"	1 $\overline{MR}$ -1Q0	"	"	"
"	"	"	151	"	"	"	"		"	OUT	"		"		"	1 $\overline{MR}$ -1Q1	"	"	"
"	"	"	152	"	"	"	"		"		"	OUT	"		"	1 $\overline{MR}$ -1Q2	"	"	"
"	"	"	153	"	"	"	"		"		"		"	OUT	"	1 $\overline{MR}$ -1Q3	"	"	"
"	"	"	154												"	2 $\overline{MR}$ -2Q0	"	"	"
"	"	"	155												"	2 $\overline{MR}$ -2Q1	"	"	"
"	"	"	156												"	2 $\overline{MR}$ -2Q2	"	"	"
"	"	"	157												"	2 $\overline{MR}$ -2Q3	"	"	"
"	"	"	158												"	2 $\overline{MR}$ -2Q0	"	"	"
"	"	"	159												"	2 $\overline{MR}$ -2Q1	"	"	"
"	"	"	160												"	2 $\overline{MR}$ -2Q2	"	"	"
"	"	"	161												"	2 $\overline{MR}$ -2Q3	"	"	"

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	13	14	15	16	17	18	19	20	21	22	23	24	Meas. terminal	Test limits		
			Test No.	2 $\overline{MR}$	2 $\overline{E}$ 0	2 $\overline{E}$ 1	2D0	2Q0	2D1	2Q1	2D2	2Q2	2D3	2Q3	V_{CC}		Min	Max	Unit
9	t_{PHL2}	3003	130	5.0 V											5.0 V	1D0-1Q0	3	28	ns
T_C = 25°C	"	Fig 6	131	"											"	1D1-1Q1	"	"	"
"	"	"	132	"											"	1D2-1Q2	"	"	"
"	"	"	133	"											"	1D3-1Q3	"	"	"
"	"	"	134	"											"	1D0-1Q0	"	"	"
"	"	"	135	"											"	1D1-1Q1	"	"	"
"	"	"	136	"											"	1D2-1Q2	"	"	"
"	"	"	137	"											"	1D3-1Q3	"	"	"
"	"	"	138	"	IN	GND	IN	OUT							"	2D0-2Q0	"	"	"
"	"	"	139	"	"	"			IN	OUT					"	2D1-2Q1	"	"	"
"	"	"	140	"	"	"					IN	OUT			"	2D2-2Q2	"	"	"
"	"	"	141	"	"	"							IN	OUT	"	2D3-2Q3	"	"	"
"	"	"	142	"	GND	IN	IN	OUT							"	2D0-2Q0	"	"	"
"	"	"	143	"	"	"			IN	OUT					"	2D1-2Q1	"	"	"
"	"	"	144	"	"	"					IN	OUT			"	2D2-2Q2	"	"	"
"	"	"	145	"	"	"							IN	OUT	"	2D3-2Q3	"	"	"
"	t_{PHL3}	3003	146												"	1 $\overline{MR}$ -1Q0	"	25	"
"	"	Fig 6-C	147												"	1 $\overline{MR}$ -1Q1	"	"	"
"	"	"	148												"	1 $\overline{MR}$ -1Q2	"	"	"
"	"	"	149												"	1 $\overline{MR}$ -1Q3	"	"	"
"	"	"	150												"	1 $\overline{MR}$ -1Q0	"	"	"
"	"	"	151												"	1 $\overline{MR}$ -1Q1	"	"	"
"	"	"	152												"	1 $\overline{MR}$ -1Q2	"	"	"
"	"	"	153												"	1 $\overline{MR}$ -1Q3	"	"	"
"	"	"	154	IN	IN	GND	5.0 V	OUT	5.0 V		5.0 V		5.0 V		"	2 $\overline{MR}$ -2Q0	"	"	"
"	"	"	155	"	"	"	"		"	OUT	"		"		"	2 $\overline{MR}$ -2Q1	"	"	"
"	"	"	156	"	"	"	"		"		"	OUT	"		"	2 $\overline{MR}$ -2Q2	"	"	"
"	"	"	157	"	"	"	"		"		"		"	OUT	"	2 $\overline{MR}$ -2Q3	"	"	"
"	"	"	158	"	GND	IN	"	OUT	"		"		"		"	2 $\overline{MR}$ -2Q0	"	"	"
"	"	"	159	"	"	"	"		"	OUT	"		"		"	2 $\overline{MR}$ -2Q1	"	"	"
"	"	"	160	"	"	"	"		"		"	OUT	"		"	2 $\overline{MR}$ -2Q2	"	"	"
"	"	"	161	"	"	"	"		"		"		"	OUT	"	2 $\overline{MR}$ -2Q3	"	"	"

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	1	2	3	4	5	6	7	8	9	10	11	12	Meas. terminal	Test limits		
			Test No.	$1\overline{MR}$	$1\overline{E}0$	$1\overline{E}1$	1D0	1Q0	1D1	1Q1	1D2	1Q2	1D3	1Q3	GND		Min	Max	Unit
10	t_{PLH1}	3003	162	5.0 V	IN	GND	IN	OUT							GND	$1\overline{E}$ 0-1Q0	3	60	ns
T_C = 125°C	"	Fig 6-B	163	"	"	"			IN	OUT					"	$1\overline{E}$ 1-1Q1	"	"	"
"	"	"	164	"	"	"					IN	OUT			"	$1\overline{E}$ 2-1Q2	"	"	"
"	"	"	165	"	"	"							IN	OUT	"	$1\overline{E}$ 3-1Q3	"	"	"
"	"	"	166	"	GND	IN	IN	OUT							"	$1\overline{E}$ 0-1Q0	"	"	"
"	"	"	167	"	"	"			IN	OUT					"	$1\overline{E}$ 1-1Q1	"	"	"
"	"	"	168	"	"	"					IN	OUT			"	$1\overline{E}$ 2-1Q2	"	"	"
"	"	"	169	"	"	"							IN	OUT	"	$1\overline{E}$ 3-1Q3	"	"	"
"	"	"	170	"											"	$2\overline{E}$ 0-2Q0	"	"	"
"	"	"	171	"											"	$2\overline{E}$ 1-2Q1	"	"	"
"	"	"	172	"											"	$2\overline{E}$ 2-2Q2	"	"	"
"	"	"	173	"											"	$2\overline{E}$ 3-2Q3	"	"	"
"	"	"	174	"											"	$2\overline{E}$ 0-2Q0	"	"	"
"	"	"	175	"											"	$2\overline{E}$ 1-2Q1	"	"	"
"	"	"	176	"											"	$2\overline{E}$ 2-2Q2	"	"	"
"	"	"	177	"											"	$2\overline{E}$ 3-2Q3	"	"	"

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	1	2	3	4	5	6	7	8	9	10	11	12	Meas. terminal	Test limits		
			Test No.	2 $\overline{MR}$	2 $\overline{E}$ 0	2 $\overline{E}$ 1	2D0	2Q0	2D1	2Q1	2D2	2Q2	2D3	2Q3	V_{CC}		Min	Max	Unit
10	t_{PLH1}	3003	162	5.0 V											5.0 V	1 $\overline{E}$ 0-1Q0	3	60	ns
T_C = 125°C	"	Fig 6-B	163	"											"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	164	"											"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	165	"											"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	166	"											"	1 $\overline{E}$ 0-1Q0	"	"	"
"	"	"	167	"											"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	168	"											"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	169	"											"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	170	"	IN	GND	IN	OUT							"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	171	"	"	"			IN	OUT					"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	172	"	"	"					IN	OUT			"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	173	"	"	"							IN	OUT	"	2 $\overline{E}$ 3-2Q3	"	"	"
"	"	"	174	"	GND	IN	IN	OUT							"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	175	"	"	"			IN	OUT					"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	176	"	"	"					IN	OUT			"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	177	"	"	"							IN	OUT	"	2 $\overline{E}$ 3-2Q3	"	"	"

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	1	2	3	4	5	6	7	8	9	10	11	12	Meas. terminal	Test limits		
			Test No.	1 $\overline{MR}$	1 $\overline{E}$ 0	1 $\overline{E}$ 1	1D0	1Q0	1D1	1Q1	1D2	1Q2	1D3	1Q3	GND		Min	Max	Unit
10	t_{PHL1}	3003	178	5.0 V	IN	GND	IN	OUT							GND	1 $\overline{E}$ 0-1Q0	3	40	ns
T_C = 125°C	"	Fig 6-A	179	"	"	"			IN	OUT					"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	180	"	"	"					IN	OUT			"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	181	"	"	"							IN	OUT	"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	182	"	GND	IN	IN	OUT							"	1 $\overline{E}$ 0-1Q0	"	"	"
"	"	"	183	"	"	"			IN	OUT					"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	184	"	"	"					IN	OUT			"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	185	"	"	"							IN	OUT	"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	186	"											"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	187	"											"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	188	"											"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	189	"											"	2 $\overline{E}$ 3-2Q3	"	"	"
"	"	"	190	"											"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	191	"											"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	192	"											"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	193	"											"	2 $\overline{E}$ 3-2Q3	"	"	"
"	t_{PLH2}	3003	194	"	IN	GND	IN	OUT							"	1D0-1Q0	"	49	"
"	"	Fig 6	195	"	"	"			IN	OUT					"	1D1-1Q1	"	"	"
"	"	"	196	"	"	"					IN	OUT			"	1D2-1Q2	"	"	"
"	"	"	197	"	"	"							IN	OUT	"	1D3-1Q3	"	"	"
"	"	"	198	"	GND	IN	IN	OUT							"	1D0-1Q0	"	"	"
"	"	"	199	"	"	"			IN	OUT					"	1D1-1Q1	"	"	"
"	"	"	200	"	"	"					IN	OUT			"	1D2-1Q2	"	"	"
"	"	"	201	"	"	"							IN	OUT	"	1D3-1Q3	"	"	"
"	"	"	202	"											"	2D0-2Q0	"	"	"
"	"	"	203	"											"	2D1-2Q1	"	"	"
"	"	"	204	"											"	2D2-2Q2	"	"	"
"	"	"	205	"											"	2D3-2Q3	"	"	"
"	"	"	206	"											"	2D0-2Q0	"	"	"
"	"	"	207	"											"	2D1-2Q1	"	"	"
"	"	"	208	"											"	2D2-2Q2	"	"	"
"	"	"	209	"											"	2D3-2Q3	"	"	"

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	13	14	15	16	17	18	19	20	21	22	23	24	Meas. terminal	Test limits		
			Test No.	2 $\overline{MR}$	2 $\overline{E}$ 0	2 $\overline{E}$ 1	2D0	2Q0	2D1	2Q1	2D2	2Q2	2D3	2Q3	V_{CC}		Min	Max	Unit
10	t_{PHL1}	3003	178	5.0 V											5.0 V	1 $\overline{E}$ 0-1Q0	3	40	ns
T_C = 125°C	"	Fig 6-A	179	"											"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	180	"											"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	181	"											"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	182	"											"	1 $\overline{E}$ 0-1Q0	"	"	"
"	"	"	183	"											"	1 $\overline{E}$ 1-1Q1	"	"	"
"	"	"	184	"											"	1 $\overline{E}$ 2-1Q2	"	"	"
"	"	"	185	"											"	1 $\overline{E}$ 3-1Q3	"	"	"
"	"	"	186	"	IN	GND	IN	OUT							"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	187	"	"	"			IN	OUT					"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	188	"	"	"					IN	OUT			"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	189	"	"	"							IN	OUT	"	2 $\overline{E}$ 3-2Q3	"	"	"
"	"	"	190	"	GND	IN	IN	OUT							"	2 $\overline{E}$ 0-2Q0	"	"	"
"	"	"	191	"	"	"			IN	OUT					"	2 $\overline{E}$ 1-2Q1	"	"	"
"	"	"	192	"	"	"					IN	OUT			"	2 $\overline{E}$ 2-2Q2	"	"	"
"	"	"	193	"	"	"							IN	OUT	"	2 $\overline{E}$ 3-2Q3	"	"	"
"	t_{PLH2}	3003	194	"											"	1D0-1Q0	"	49	"
"	"	Fig 6	195	"											"	1D1-1Q1	"	"	"
"	"	"	196	"											"	1D2-1Q2	"	"	"
"	"	"	197	"											"	1D3-1Q3	"	"	"
"	"	"	198	"											"	1D0-1Q0	"	"	"
"	"	"	199	"											"	1D1-1Q1	"	"	"
"	"	"	200	"											"	1D2-1Q2	"	"	"
"	"	"	201	"											"	1D3-1Q3	"	"	"
"	"	"	202	"	IN	GND	IN	OUT							"	2D0-2Q0	"	"	"
"	"	"	203	"	"	"			IN	OUT					"	2D1-2Q1	"	"	"
"	"	"	204	"	"	"					IN	OUT			"	2D2-2Q2	"	"	"
"	"	"	205	"	"	"							IN	OUT	"	2D3-2Q3	"	"	"
"	"	"	206	"	GND	IN	IN	OUT							"	2D0-2Q0	"	"	"
"	"	"	207	"	"	"			IN	OUT					"	2D1-2Q1	"	"	"
"	"	"	208	"	"	"					IN	OUT			"	2D2-2Q2	"	"	"
"	"	"	209	"	"	"							IN	OUT	"	2D3-2Q3	"	"	"

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	1	2	3	4	5	6	7	8	9	10	11	12	Meas. terminal	Test limits		
			Test No.	1 $\overline{MR}$	1 $\overline{E}$ 0	1 $\overline{E}$ 1	1D0	1Q0	1D1	1Q1	1D2	1Q2	1D3	1Q3	GND		Min	Max	Unit
10	t_{PHL2}	3003	210	5.0 V	IN	GND	IN	OUT							GND	1D0-1Q0	3	37	ns
T_C = 125°C	"	Fig 6	211	"	"	"			IN	OUT					"	1D1-1Q1	"	"	"
"	"	"	212	"	"	"					IN	OUT			"	1D2-1Q2	"	"	"
"	"	"	213	"	"	"							IN	OUT	"	1D3-1Q3	"	"	"
"	"	"	214	"	GND	IN	IN	OUT							"	1D0-1Q0	"	"	"
"	"	"	215	"	"	"			IN	OUT					"	1D1-1Q1	"	"	"
"	"	"	216	"	"	"					IN	OUT			"	1D2-1Q2	"	"	"
"	"	"	217	"	"	"							IN	OUT	"	1D3-1Q3	"	"	"
"	"	"	218	"											"	2D0-2Q0	"	"	"
"	"	"	219	"											"	2D1-2Q1	"	"	"
"	"	"	220	"											"	2D2-2Q2	"	"	"
"	"	"	221	"											"	2D3-2Q3	"	"	"
"	"	"	222	"											"	2D0-2Q0	"	"	"
"	"	"	223	"											"	2D1-2Q1	"	"	"
"	"	"	224	"											"	2D2-2Q2	"	"	"
"	"	"	225	"											"	2D3-2Q3	"	"	"
"	t_{PHL3}	3003	226	IN	IN	GND	5.0 V	OUT	5.0 V		5.0 V		5.0 V		"	1 $\overline{MR}$ -1Q0	"	"	"
"	"	Fig 6-C	227	"	"	"	"		"	OUT	"		"		"	1 $\overline{MR}$ -1Q1	"	"	"
"	"	"	228	"	"	"	"		"		"	OUT	"		"	1 $\overline{MR}$ -1Q2	"	"	"
"	"	"	229	"	"	"	"		"		"		"	OUT	"	1 $\overline{MR}$ -1Q3	"	"	"
"	"	"	230	"	GND	IN	"	OUT	"		"		"		"	1 $\overline{MR}$ -1Q0	"	"	"
"	"	"	231	"	"	"	"		"	OUT	"		"		"	1 $\overline{MR}$ -1Q1	"	"	"
"	"	"	232	"	"	"	"		"		"	OUT	"		"	1 $\overline{MR}$ -1Q2	"	"	"
"	"	"	233	"	"	"	"		"		"		"	OUT	"	1 $\overline{MR}$ -1Q3	"	"	"
"	"	"	234												"	2 $\overline{MR}$ -2Q0	"	"	"
"	"	"	235												"	2 $\overline{MR}$ -2Q1	"	"	"
"	"	"	236												"	2 $\overline{MR}$ -2Q2	"	"	"
"	"	"	237												"	2 $\overline{MR}$ -2Q3	"	"	"
"	"	"	238												"	2 $\overline{MR}$ -2Q0	"	"	"
"	"	"	239												"	2 $\overline{MR}$ -2Q1	"	"	"
"	"	"	240												"	2 $\overline{MR}$ -2Q2	"	"	"
"	"	"	241												"	2 $\overline{MR}$ -2Q3	"	"	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																		

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case J, K, L, Z	13	14	15	16	17	18	19	20	21	22	23	24	Meas. terminal	Test limits		
			Test No.	2 $\overline{MR}$	2 $\overline{E}$ 0	2 $\overline{E}$ 1	2D0	2Q0	2D1	2Q1	2D2	2Q2	2D3	2Q3	V_{CC}		Min	Max	Unit
10	t_{PHL2}	3003	210	5.0 V											5.0 V	1D0-1Q0	3	37	ns
T_C = 125°C	"	Fig 6	211	"											"	1D1-1Q1	"	"	"
"	"	"	212	"											"	1D2-1Q2	"	"	"
"	"	"	213	"											"	1D3-1Q3	"	"	"
"	"	"	214	"											"	1D0-1Q0	"	"	"
"	"	"	215	"											"	1D1-1Q1	"	"	"
"	"	"	216	"											"	1D2-1Q2	"	"	"
"	"	"	217	"											"	1D3-1Q3	"	"	"
"	"	"	218	"	IN	GND	IN	OUT							"	2D0-2Q0	"	"	"
"	"	"	219	"	"	"			IN	OUT					"	2D1-2Q1	"	"	"
"	"	"	220	"	"	"					IN	OUT			"	2D2-2Q2	"	"	"
"	"	"	221	"	"	"							IN	OUT	"	2D3-2Q3	"	"	"
"	"	"	222	"	GND	IN	IN	OUT							"	2D0-2Q0	"	"	"
"	"	"	223	"	"	"			IN	OUT					"	2D1-2Q1	"	"	"
"	"	"	224	"	"	"					IN	OUT			"	2D2-2Q2	"	"	"
"	"	"	225	"	"	"							IN	OUT	"	2D3-2Q3	"	"	"
"	t_{PHL3}	3003	226												"	1 $\overline{MR}$ -1Q0	"	"	"
"	"	Fig 6-C	227												"	1 $\overline{MR}$ -1Q1	"	"	"
"	"	"	228												"	1 $\overline{MR}$ -1Q2	"	"	"
"	"	"	229												"	1 $\overline{MR}$ -1Q3	"	"	"
"	"	"	230												"	1 $\overline{MR}$ -1Q0	"	"	"
"	"	"	231												"	1 $\overline{MR}$ -1Q1	"	"	"
"	"	"	232												"	1 $\overline{MR}$ -1Q2	"	"	"
"	"	"	233												"	1 $\overline{MR}$ -1Q3	"	"	"
"	"	"	234	IN	IN	GND	5.0 V	OUT	5.0 V		5.0 V		5.0 V		"	2 $\overline{MR}$ -2Q0	"	"	"
"	"	"	235	"	"	"	"		"	OUT	"		"		"	2 $\overline{MR}$ -2Q1	"	"	"
"	"	"	236	"	"	"	"		"		"	OUT	"		"	2 $\overline{MR}$ -2Q2	"	"	"
"	"	"	237	"	"	"	"		"		"		"	OUT	"	2 $\overline{MR}$ -2Q3	"	"	"
"	"	"	238	"	GND	IN	"	OUT	"		"		"		"	2 $\overline{MR}$ -2Q0	"	"	"
"	"	"	239	"	"	"	"		"	OUT	"		"		"	2 $\overline{MR}$ -2Q1	"	"	"
"	"	"	240	"	"	"	"		"		"	OUT	"		"	2 $\overline{MR}$ -2Q2	"	"	"
"	"	"	241	"	"	"	"		"		"		"	OUT	"	2 $\overline{MR}$ -2Q3	"	"	"
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																		

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	$\overline{E}$	$\overline{S}$ 0	D0	D1	$\overline{S}$ 2	D2	D3	GND	$\overline{MR}$	Q3	$\overline{S}$ 3	Q2	Q1	$\overline{S}$ 1	Q0	V_{CC}		Min	Max	Unit
1	V_{OH}	3006	1	0.8 V	0.8 V	2.0 V	2.0 V	0.8 V	2.0 V	2.0 V	GND	2.0 V		0.8 V			0.8 V	-800µA	4.5 V	Q0	2.4		V
T_C = 25°C	"	"	2	"	"	"	"	"	"	"	"	"		"		-800µA	"		"	Q1	"		"
"	"	"	3	"	"	"	"	"	"	"	"	"		"	-800µA		"		"	Q2	"		"
"	"	"	4	"	"	"	"	"	"	"	"	"	-800µA	"			"		"	Q3	"		"
"	V_{OL}	3007	5	"	"	0.8 V	0.8 V	"	0.8 V	0.8 V	"	"		"			"	16 mA	"	Q0		0.4	"
"	"	"	6	"	"	"	"	"	"	"	"	"		"		16 mA	"		"	Q1		"	"
"	"	"	7	"	"	"	"	"	"	"	"	"		"	16 mA		"		"	Q2		"	"
"	"	"	8	"	"	"	"	"	"	"	"	"	16 mA	"			"		"	Q3		"	"
"	V_{IC}		9	-12 mA							"								"	$\overline{E}$		-1.5	"
"	"		10		-12 mA						"								"	$\overline{S}$ 0		"	"
"	"		11			-12 mA					"								"	D0		"	"
"	"		12				-12 mA				"								"	D1		"	"
"	"		13					-12 mA			"								"	$\overline{S}$ 2		"	"
"	"		14						-12 mA		"								"	D2		"	"
"	"		15							-12 mA	"								"	D3		"	"
"	"		16								"	-12 mA							"	$\overline{MR}$		"	"
"	"		17								"			-12 mA					"	$\overline{S}$ 3		"	"
"	"		18								"						-12 mA		"	$\overline{S}$ 1		"	"
"	I_{IL1}	3009	19	GND	GND	0.4 V		GND			"			GND			GND		5.5 V	D0	-0.7	-2.7	mA
"	"	"	20	"	"		0.4 V	"			"			"			"		"	D1	"	"	"
"	"	"	21	"	"			"	0.4 V		"			"			"		"	D2	"	"	"
"	"	"	22	"	"			"		0.4 V	"			"			"		"	D3	"	"	"
"	I_{IL2}	"	23	0.4 V							"								"	$\overline{E}$	"	-1.6	"
"	"	"	24		0.4 V						"								"	$\overline{S}$ 0	"	"	"
"	"	"	25					0.4 V			"								"	$\overline{S}$ 2	"	"	"
"	"	"	26								"	0.4 V							"	$\overline{MR}$	"	"	"
"	"	"	27								"			0.4 V					"	$\overline{S}$ 3	"	"	
"	"	"	28								"						0.4 V		"	$\overline{S}$ 1	"	"	"
"	I_{IH1}	3010	29	5.5 V	5.5 V	2.4 V		5.5 V			"	GND		5.5 V			5.5 V		"	D0		80	µA
"	"	"	30	"	"		2.4 V	"			"	"		"			"		"	D1		"	"
"	"	"	31	"	"			"	2.4 V		"	"		"			"		"	D2		"	"
"	"	"	32	"	"			"		2.4 V	"	"		"			"		"	D3		"	"

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TABLE III. Group A inspection for device type 04.

Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1 $\overline{E}$	2 $\overline{S}$ 0	3 D0	4 D1	5 $\overline{S}$ 2	6 D2	7 D3	8 GND	9 $\overline{MR}$	10 Q3	11 $\overline{S}$ 3	12 Q2	13 Q1	14 $\overline{S}$ 1	15 Q0	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1	I_{IH2}	3010	33	5.5 V	5.5 V	5.5 V		5.5 V			GND	GND		5.5 V			5.5 V		5.5 V	D0		200	µA
T_C = 25°C	"	"	34	"	"		5.5 V	"			"	"		"			"		"	D1		"	"
"	"	"	35	"	"			"	5.5 V		"	"		"			"		"	D2		"	"
"	"	"	36	"	"			"		5.5 V	"	"		"			"		"	D3		"	"
"	I_{IH3}	3006	37	2.4 V							"								"	$\overline{E}$		40	"
"	"	"	38		2.4 V						"								"	$\overline{S}$ 0		"	"
"	"	"	39					2.4 V			"								"	$\overline{S}$ 2		"	"
"	"	"	40								"	2.4 V							"	$\overline{MR}$		"	"
"	"	"	41								"			2.4 V					"	$\overline{S}$ 3		"	"
"	"	"	42								"						2.4 V		"	$\overline{S}$ 1		"	"
"	I_{IH4}	"	43	5.5 V							"								"	$\overline{E}$		100	"
"	"	"	44		5.5 V						"								"	$\overline{S}$ 0		"	"
"	"	"	45					5.5 V			"								"	$\overline{S}$ 2		"	"
"	"	"	46								"	5.5 V							"	$\overline{MR}$		"	"
"	"	"	47								"			5.5 V					"	$\overline{S}$ 3		"	"
"	"	"	48								"						5.5 V		"	$\overline{S}$ 1		"	"
"	I_{OS}	3011	49	GND	GND	4.5 V					"	5.5 V				GND		GND	"	Q0	-20	-100	mA
"	"	"	50	"			4.5 V				"	"				GND	GND		"	Q1	"	"	"
"	"	"	51	"				GND	4.5 V		"	"			GND				"	Q2	"	"	"
"	"	"	52	"						4.5 V	"	"	GND	GND					"	Q3	"	"	"
"	I_{CC}	3005	53	4.5 V	GND			GND			"	GND		GND			GND		"	V_{CC}		55	"
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
9	t_{PLH1}	3003	54	IN	GND	IN					GND	5.0 V						OUT	5.0 V	$\overline{E}$ -Q0	9	30	ns
T_C = 25°C	"	Fig 7-A	55	"			IN				"	"				OUT	GND		"	$\overline{E}$ -Q1	"	"	"
"	"	"	56	"				GND	IN		"	"			OUT				"	$\overline{E}$ -Q2	"	"	"
"	"	"	57	"						IN	"	"	OUT	GND					"	$\overline{E}$ -Q3	"	"	"
"	t_{PHL1}	3003	58	IN	GND	IN					"	"						OUT	"	$\overline{E}$ -Q0	"	29	ns
"	"	Fig 7-B	59	"			IN				"	"				OUT	GND		"	$\overline{E}$ -Q1	"	"	"
"	"	"	60	"				GND	IN		"	"			OUT				"	$\overline{E}$ -Q2	"	"	"
"	"	"	61	"						IN	"	"	OUT	GND					"	$\overline{E}$ -Q3	"	"	"

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TABLE III. Group A inspection for device type 04.

Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	$\overline{E}$	$\overline{S}$ 0	D0	D1	$\overline{S}$ 2	D2	D3	GND	$\overline{MR}$	Q3	$\overline{S}$ 3	Q2	Q1	$\overline{S}$ 1	Q0	V_{CC}		Min	Max	Unit
9	t_{PHL1}	3003	62	IN	GND	IN					GND	5.0 V						OUT	5.0 V	$\overline{E}$ -Q0	9	29	ns
T_C = 25°C	"	Fig 7-C	63	"			IN				"	"				OUT	GND		"	$\overline{E}$ -Q1	"	"	"
"	"	"	64	"				GND	IN		"	"			OUT				"	$\overline{E}$ -Q2	"	"	"
"	"	"	65	"						IN	"	"	OUT	GND					"	$\overline{E}$ -Q3	"	"	"
"	t_{PLH2}	3003	66	"	GND	IN					"	"						OUT	"	D0-Q0	"	21	ns
"	"	Fig 7-B	67	"			IN				"	"				OUT	GND		"	D1-Q1	"	"	"
"	"	"	68	"				GND	IN		"	"			OUT				"	D2-Q2	"	"	"
"	"	"	69	"						IN	"	"	OUT	GND					"	D3-Q3	"	"	"
"	t_{PLH2}	3003	70	"	GND	IN					"	"						OUT	"	D0-Q0	"	"	"
"	"	Fig 7-C	71	"			IN				"	"				OUT	GND		"	D1-Q1	"	"	"
"	"	"	72	"				GND	IN		"	"			OUT				"	D2-Q2	"	"	"
"	"	"	73	"						IN	"	"	OUT	GND					"	D3-Q3	"	"	"
"	t_{PLH2}	3003	74	GND	GND	IN					"	"						OUT	"	D0-Q0	"	"	"
"	"	Fig 7-D	75	"			IN				"	"				OUT	GND		"	D1-Q1	"	"	"
"	"	"	76	"				GND	IN		"	"			OUT				"	D2-Q2	"	"	"
"	"	"	77	"						IN	"	"	OUT	GND					"	D3-Q3	"	"	"
"	t_{PHL2}	3003	78	"	GND	IN					"	"						OUT	"	D0-Q0	3	30	"
"	"	Fig 7-E	79	"			IN				"	"				OUT	GND		"	D1-Q1	"	"	"
"	"	"	80	"				GND	IN		"	"			OUT				"	D2-Q2	"	"	"
"	"	"	81	"						IN	"	"	OUT	GND					"	D3-Q3	"	"	"
"	t_{PHL2}	3003	82	IN	GND	IN					"	"						OUT	"	D0-Q0	"	"	"
"	"	Fig 7-A	83	"			IN				"	"				OUT	GND		"	D1-Q1	"	"	"
"	"	"	84	"				GND	IN		"	"			OUT				"	D2-Q2	"	"	"
"	"	"	85	"						IN	"	"	OUT	GND					"	D3-Q3	"	"	"
"	t_{PHL3}	3003	86	"	GND	5.0 V					"	IN						OUT	"	$\overline{MR}$ -Q0	9	23	"
"	"	Fig 7-F	87	"			5.0 V				"	"				OUT	GND		"	$\overline{MR}$ -Q1	"	"	"
"	"	"	88	"				GND	5.0 V		"	"			OUT				"	$\overline{MR}$ -Q2	"	"	"
"	"	"	89	"						5.0 V	"	"	OUT	GND					"	$\overline{MR}$ -Q3	"	"	"
"	t_{PLH4}	3003	90	GND	IN	IN					"	5.0 V						OUT	"	$\overline{S}$ 0-Q0	"	29	"
"	"	Fig 7-E	91	"			IN				"	"				OUT	IN		"	$\overline{S}$ 1-Q1	"	"	"
"	"	"	92	"				IN	IN		"	"			OUT				"	$\overline{S}$ 2-Q2	"	"	"
"	"	"	93	"						IN	"	"	OUT	IN					"	$\overline{S}$ 3-Q3	"	"	"

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TABLE III. Group A inspection for device type 04.

Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	$\overline{E}$	$\overline{S}$ 0	D0	D1	$\overline{S}$ 2	D2	D3	GND	$\overline{MR}$	Q3	$\overline{S}$ 3	Q2	Q1	$\overline{S}$ 1	Q0	V_{CC}		Min	Max	Unit
10	t_{PLH1}	3003	94	IN	GND	IN					GND	5.0 V						OUT	5.0 V	$\overline{E}$ -Q0	9	47	ns
T_C = 125°C	"	Fig 7-A	95	"			IN				"	"				OUT	GND		"	$\overline{E}$ -Q1	"	"	"
"	"	"	96	"				GND	IN		"	"			OUT				"	$\overline{E}$ -Q2	"	"	"
"	"	"	97	"						IN	"	"	OUT	GND					"	$\overline{E}$ -Q3	"	"	"
"	t_{PHL1}	3003	98	"	GND	IN					"	"						OUT	"	$\overline{E}$ -Q0	"	33	ns
"	"	Fig 7-B	99	"			IN				"	"				OUT	GND		"	$\overline{E}$ -Q1	"	"	"
"	"	"	100	"				GND	IN		"	"			OUT				"	$\overline{E}$ -Q2	"	"	"
"	"	"	101	"						IN	"	"	OUT	GND					"	$\overline{E}$ -Q3	"	"	"
"	t_{PHL1}	3003	102	"	GND	IN					"	"						OUT	"	$\overline{E}$ -Q0	"	"	ns
"	"	Fig 7-C	103	"			IN				"	"				OUT	GND		"	$\overline{E}$ -Q1	"	"	"
"	"	"	104	"				GND	IN		"	"			OUT				"	$\overline{E}$ -Q2	"	"	"
"	"	"	105	"						IN	"	"	OUT	GND					"	$\overline{E}$ -Q3	"	"	"
"	t_{PLH2}	3003	106	"	GND	IN					"	"						OUT	"	D0-Q0	"	37	ns
"	"	Fig 7-B	107	"			IN				"	"				OUT	GND		"	D1-Q1	"	"	"
"	"	"	108	"				GND	IN		"	"			OUT				"	D2-Q2	"	"	"
"	"	"	109	"						IN	"	"	OUT	GND					"	D3-Q3	"	"	"
"	t_{PLH2}	3003	110	"	GND	IN					"	"						OUT	"	D0-Q0	"	"	"
"	"	Fig 7-C	111	"			IN				"	"				OUT	GND		"	D1-Q1	"	"	"
"	"	"	112	"				GND	IN		"	"			OUT				"	D2-Q2	"	"	"
"	"	"	113	"						IN	"	"	OUT	GND					"	D3-Q3	"	"	"
"	t_{PLH2}	3003	114	GND	GND	IN					"	"						OUT	"	D0-Q0	"	"	"
"	"	Fig 7-D	115	"			IN				"	"				OUT	GND		"	D1-Q1	"	"	"
"	"	"	116	"				GND	IN		"	"			OUT				"	D2-Q2	"	"	"
"	"	"	117	"						IN	"	"	OUT	GND					"	D3-Q3	"	"	"
"	t_{PHL2}	3003	118	"	GND	IN					"	"						OUT	"	D0-Q0	3	"	"
"	"	Fig 7-E	119	"			IN				"	"				OUT	GND		"	D1-Q1	"	"	"
"	"	"	120	"				GND	IN		"	"			OUT				"	D2-Q2	"	"	"
"	"	"	121	"						IN	"	"	OUT	GND					"	D3-Q3	"	"	"
"	t_{PHL2}	3003	122	IN	GND	IN					"	"						OUT	"	D0-Q0	"	"	"
"	"	Fig 7-A	123	"			IN				"	"				OUT	GND		"	D1-Q1	"	"	"
"	"	"	124	"				GND	IN		"	"			OUT				"	D2-Q2	"	"	"
"	"	"	125	"						IN	"	"	OUT	GND					"	D3-Q3	"	"	"

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be H ≥ 2.0 V or L ≤ 0.8 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	$\overline{E}$	$\overline{S}$ 0	D0	D1	$\overline{S}$ 2	D2	D3	GND	$\overline{MR}$	Q3	$\overline{S}$ 3	Q2	Q1	$\overline{S}$ 1	Q0	V_{CC}		Min	Max	Unit
10	t_{PHL3}	3003	126	IN	GND	5.0 V					GND	IN						OUT	5.0 V	$\overline{MR}$ -Q0	9	30	ns
T_C = 125°C	"	Fig 7-F	127	"			5.0 V				"	"				OUT	GND		"	$\overline{MR}$ -Q1	"	"	"
"	"	"	128	"				GND	5.0 V		"	"			OUT				"	$\overline{MR}$ -Q2	"	"	"
"	"	"	129	"						5.0 V	"	"	OUT	GND					"	$\overline{MR}$ -Q3	"	"	"
"	t_{PLH4}	3003	130	GND	IN	IN					"	5.0 V						OUT	"	$\overline{S}$ 0-Q0	"	47	"
"	"	Fig 7-E	131	"			IN				"	"				OUT	IN		"	$\overline{S}$ 1-Q1	"	"	"
"	"	"	132	"				IN	IN		"	"			OUT				"	$\overline{S}$ 2-Q2	"	"	"
"	"	"	133	"						IN	"	"	OUT	IN					"	$\overline{S}$ 3-Q3	"	"	"
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																						

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5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but it is not mandatory)

6.1 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. PIN and compliance identifier, if applicable (see 1.2).
c. Requirements for delivery of one copy of the conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
d. Requirement for certificate of compliance, if applicable.
e. Requirements for notification of change of product or process to acquiring activity in addition to notification to the qualifying activity, if applicable.
f. Requirements for failure analysis (including required test condition of method 5003), corrective action and reporting of results, if applicable.
g. Requirements for product assurance options.
h. Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.
i. Requirements for "JAN" marking.
j. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43123-1199.

6.4 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.

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6.5 Abbreviations, symbols and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535 and MIL-HDBK-1331, and as follows:

GND ..... Electrical ground (common terminal)
V_{IN} ..... Voltage level at an input terminal

6.6 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish A (see 3.4). Longer lead lengths and lead forming shall not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Device type	Commercial type
01	5475, 7475 (circuit A)
01	SN5475, SN7475 (circuit B)
02	5477, 7477 (circuit A)
02	SN5477, SN7477 (circuit B)
03	9308 (circuit A)
03	54116, 74116 and SN74116 (circuit B)
03	S54116 (circuit C)
04	9314 (circuit A)
04	AM9314 (circuit B)

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5962-2104)

Review activities:
Army - MI, SM
Navy - AS, CG, MC, SH, TD
Air Force - 03, 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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